MIL-A-27328 (USAF)
14 January 1960

MILITARY SPECIFICATION

AMPLIFIER-INDICATOR GROUP, ALTITUDE-VERTICAL SPEED A/A24G-7

1. SCOPE

1.1 This specification establishes the requirements for one type of aircraft amplifier-indicator group, altitude vertical speed designated A/A24G-7.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue if effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-E-7894	Electric Power, Aircraft, Characteristics Of
MIL-P-7936	Parts And Equipment, Aeronautical, Preparation For Delivery
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports, Preparation Of
MIL-I-25467	Lighting, Integral, Instrument, General Specification For
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-D-70327	Drawings, Engineering and Associated Lists

FSC 6610

AF Hoover-Dayton Co., 23 May 1962 600

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS33586	Metals, Definition of Dissimilar

PUBLICATION

Air Force-Navy Aeronautical Bulletin

143	Specifications and Standards; Use of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 COMPONENTS.- The amplifier-indicator group shall consist of

Indicator, Altitude-Vertical Speed AAU-1/A24G-7
Amplifier, Electronic Control ASK-6/A24G-7

3.3 GENERAL SPECIFICATION.- The requirements of MIL-E-5400 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shallgovern.

3.4 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected according to ANA Bulletin 143 except as provided in 3.4.1 and 3.4.2.

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3.4.1 COMMERCIAL PARTS.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractors drawings,

3.4.2 STANDARD PARTS.- With the exception in 3.4, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.5 MATERIALS

3.5.1 FUNGUS PROOF MATERIALS.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not harmetically sealed, they shall be treated with a fungicidal agent. If used within a hermetically sealed enclosure, fungicidal treatment is not required.

3.5.2 METALS.- Metals shall be of corrosion resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

3.5.2.1 DISSIMILAR METALS.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.5.3 NONMAGNETIC MATERIALS.- Nonmagnetic materials shall be used for all parts of the instrument, except where magnetic materials are essential.

3.5.4 PROTECTIVE TREATMENT.- Materials which are utilized within the hermetically sealed case of the indicator and amplifier are considered suitably protected against corrosion.

3.5.5 TOXIC AND CORROSIVE FUMES.- The materials, as installed and under service conditions specified, shall not liberate gases or fumes which will result in corrosion such as to cause malfunction of equipment, discoloration of dials or indicia, nor shall toxic gases or fumes which are detrimental to performance of the aircraft or health of personnel, be liberated under the service conditions specified in MIL-E-5400.

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3.6 DESIGN AND CONSTRUCTION

3.6.1 GENERAL.- The indicator and amplifier specified as components in 3.2 shall be designed and manufactured in accordance with figures 1 through 7 for satisfactory performance, with maximum reliability consistent with requirements and minimum weight as specified herein.

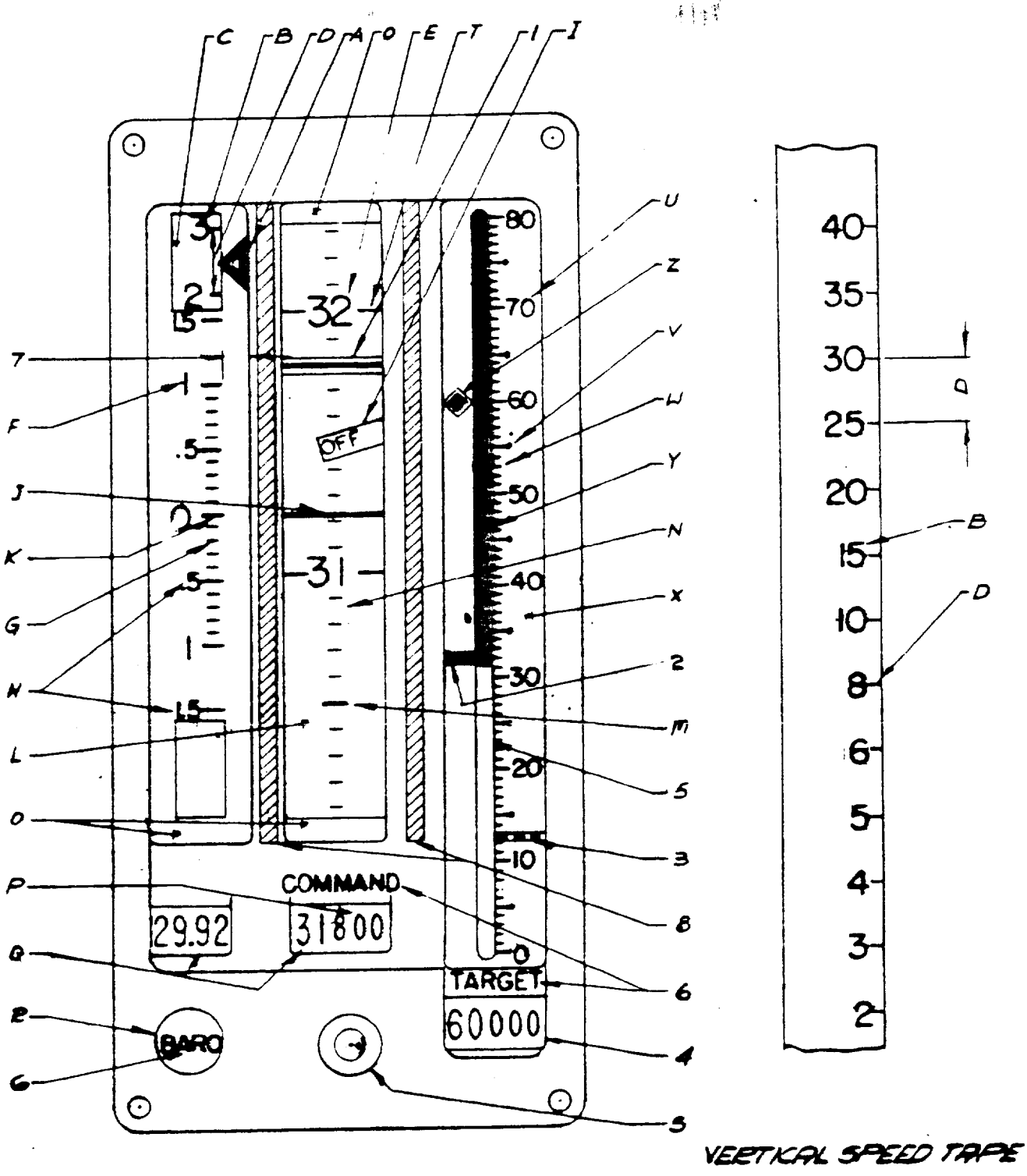
3.6.2 CASE.- The design of the case for the indicator and amplifier shall be in accordance with detailed dimensional requirements and shall provide a hermetically sealed enclosure for the internal mechanism. The design of the case shall be such that the internal mechanism may be removed from the case, replaced and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.6.2.1 FILLING MEDIUM.- The filling medium shall be of at least 98 percent purity, free of dust particles and contain not more than 0.006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium shall be approximately 1/2 to 1 atmosphere as required. Where practicable the 100 percent helium shall be utilized.

3.6.3 POWER REQUIREMENTS.- The system shall be designed to meet all applicable requirements of MIL-E-7894 and shall operate from the power defined therein. The power requirements shall be of the following type and shall not exceed the indicated amounts:

Type I AC power
Operating limits: 102 to 124V
Standby voltage limits or emergency supply: 90 to 130V
Operating frequency limits: 380 to 420 cps
Standby frequency limits or emergency supply: 380 to 420 cps
Maximum operating volt-amperes: 100
Maximum warmup time: One minute if temperature has been at -35°C or highter. Three minutes if temperature has been at -55°C.

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VERTICAL SPEED TAPE

ALTITUDE-VERTICAL SPEED

FIGURE 1.

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A - TRIANGULAR INDEX MOVES UNDER FIXED REFERENCE. VERTICAL BASE IS 3/8" LONG AND CHORD IS 7/32". BLACK CENTER WITH 1/16" WHITE BORDER.
B - 3/16" WHITE NUMERALS.
C - BLACK-DURABLE DULL.
D - BLACK ON BLACK.
E - 7/32" HIGH WHITE NUMERALS.
F - 3/16" WHITE NUMERALS.
G - 3/32" WHITE.
H - 1/8" HIGH WHITE NUMERALS.
I - RED 3/16" x 9/16" BLACK LETTERS.
J - .030 WHITE.
K - 5/32" WHITE.
L - BLACK TAPE.
M - 53/64 WHITE.
N - 3/32" WHITE
O - BLACK.
P - 7/32" WHITE NUMERALS.
Q - 1/4" WHITE NUMERALS.
R - BAKO SET KNOB - 7/8" LONG x 1/2" DIAMETER, GRAY #36231 FED. STD. 595, WITH BLACK FACE.
S - SPRING CENTERED SWITCH, GRAY #36231 FED. STD. 595.
T - 3/16" WHITE.
U - 1/8" WHITE NUMERALS.
V - 3/64" DIA. WHITE DOT.
W - 1/16" WHITE.
X - BLACK.
Y - BLACK BACKGROUND.
Z - 5/32" x 5/32" OVERALL WITH 1/32" WHITE BORDER.

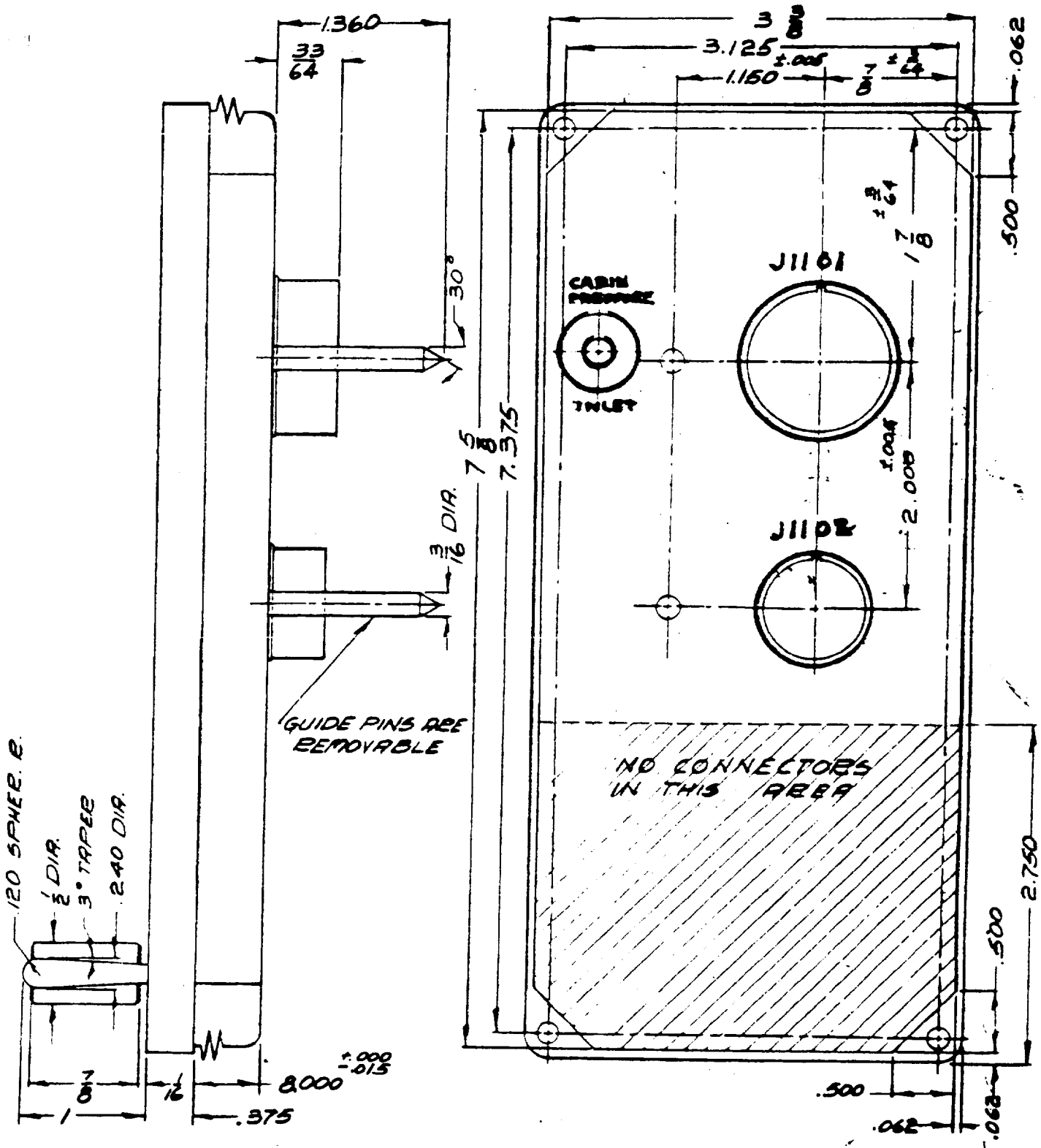
1 - 1/8" TOTAL WIDTH WITH .030 BLACK CENTER LINE WHITE.
2 - 3/32" TOTAL WIDTH 1/4" LONG WITH .030 BLACK CENTER LINE WHITE EDGES.
3 - ALTERNATE BLACK AND WHITE DASHES. BLACK STARTING AT EDGE OF THERMOMETER. 1/16" HIGH AND 1/16" WIDE. WHITE 1/16" HIGH AND 1/16" WIDE.
4 - 1/4" WHITE NUMERALS. FOR FIRST TWO NUMERALS BALANCE 7/32".
5 - 5/32" WIDE WHITE.
6 - 1/8" WHITE.
7 - GRADUATIONS 3/16" FROM EDGE.
8 - GRAY #36118 FED. STD. 595. 1/8 WIDE.

NOTES: -

-RED-FLUORESCENT RED INSTRUMENT LACQUER
#116-1548 (FROM SWITZER BROS.)
(CLEVELAND, OHIO)
-PAINT PER FEDERAL STANDARD #595
BLACK - #37038
WHITE - #37875
-STROKE WIDTH OF NUMERALS AND GRADUATIONS 1/6 to 1/8 OF HEIGHT.
-WIDTH OF NUMERALS TO BE 3/5 OF HEIGHT.

FIGURE 2. NOMENCLATURE LIST

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CASE, ALTITUDE VERTICAL SPEED
REAR & SIDE VIEW

TOLERANCES:
FRACTIONS ±1/64
DECIMALS ±.010

FIGURE 3.

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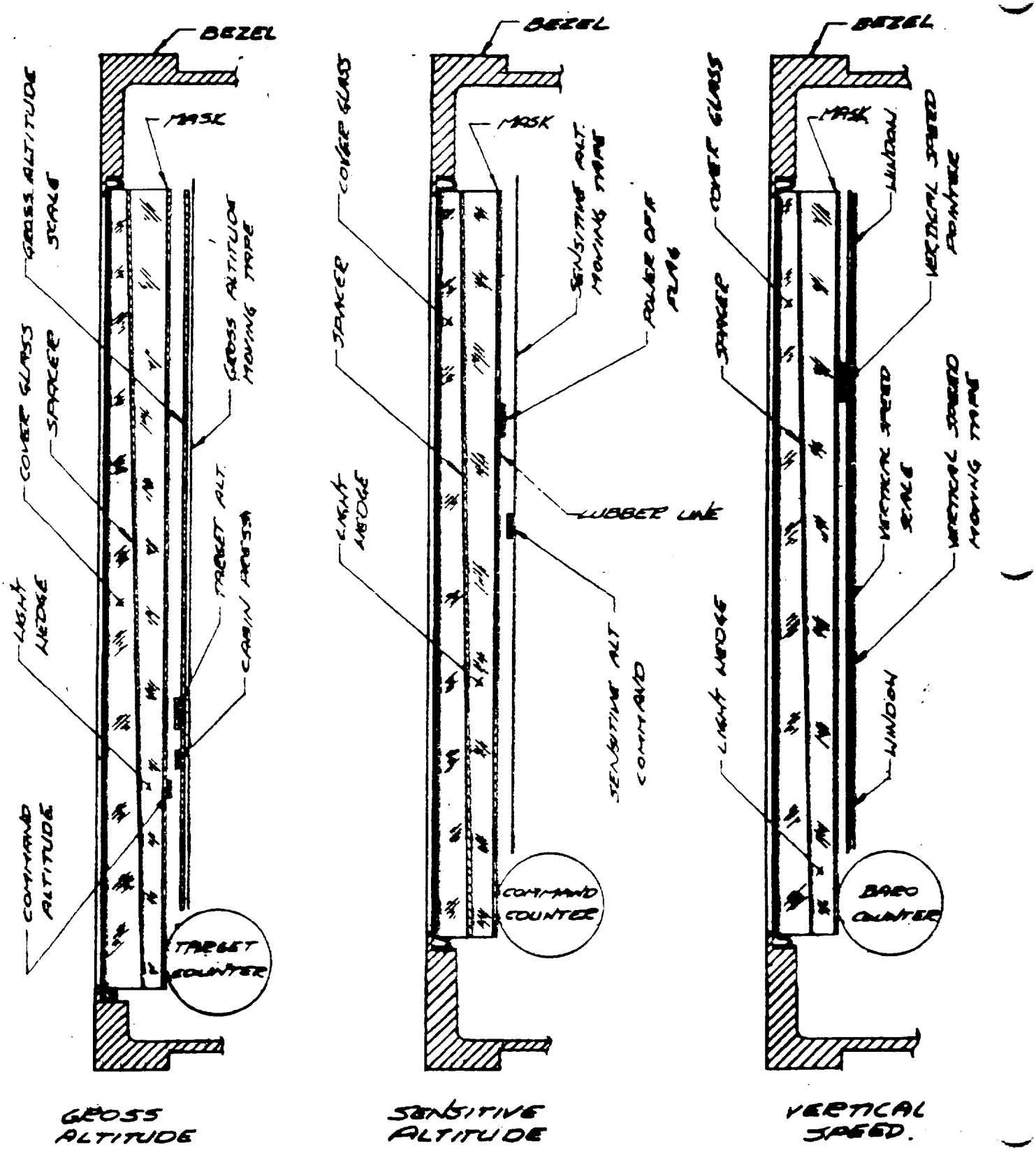
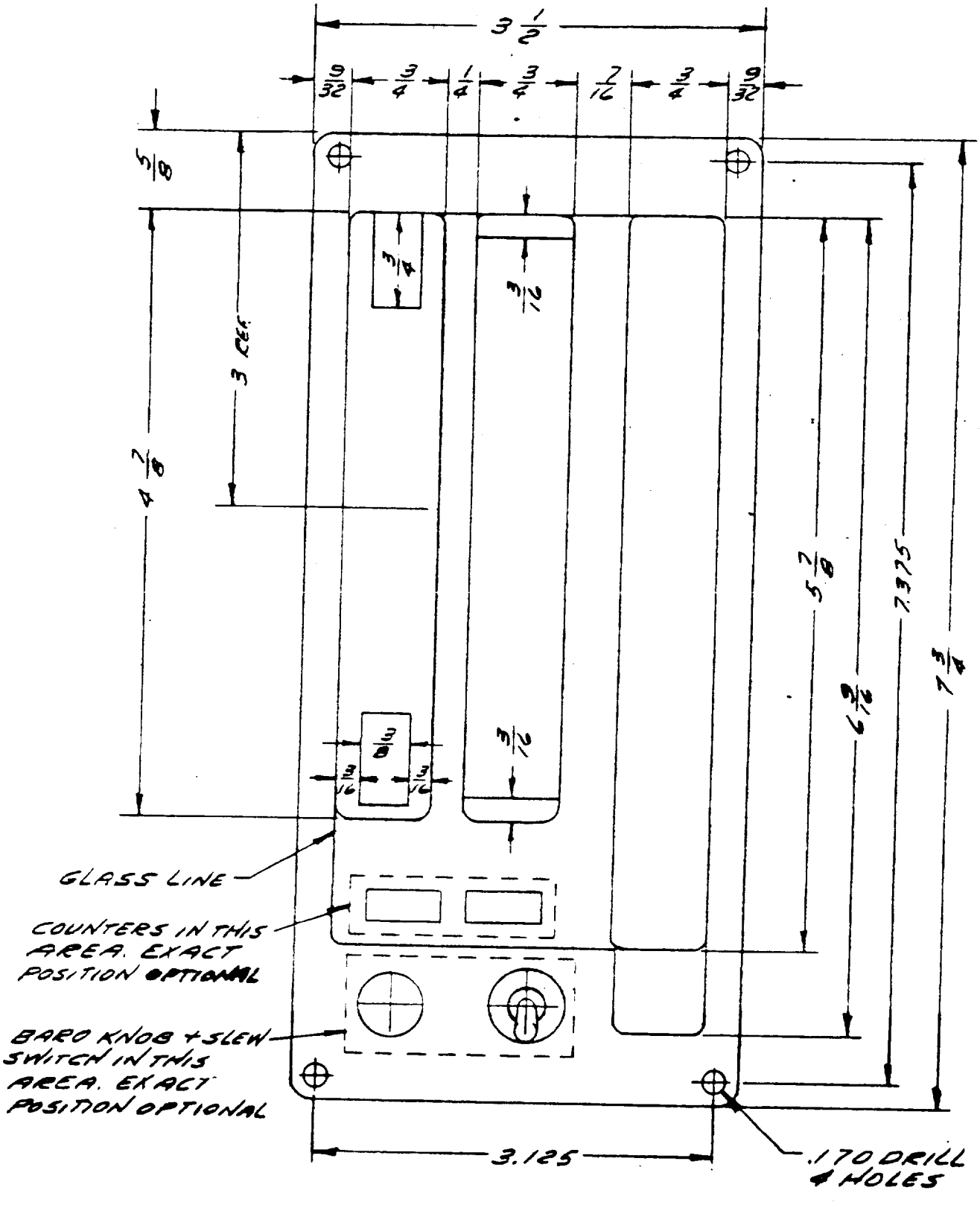


FIGURE 4. SECTIONS, ALTITUDE, VERTICAL SPEED

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TOLERANCES:
FRACTIONS ± 1/32
DECIMALS ± .010

MASK & BEZEL, ALTITUDE-VERTICAL SPEED

FIGURE 5.

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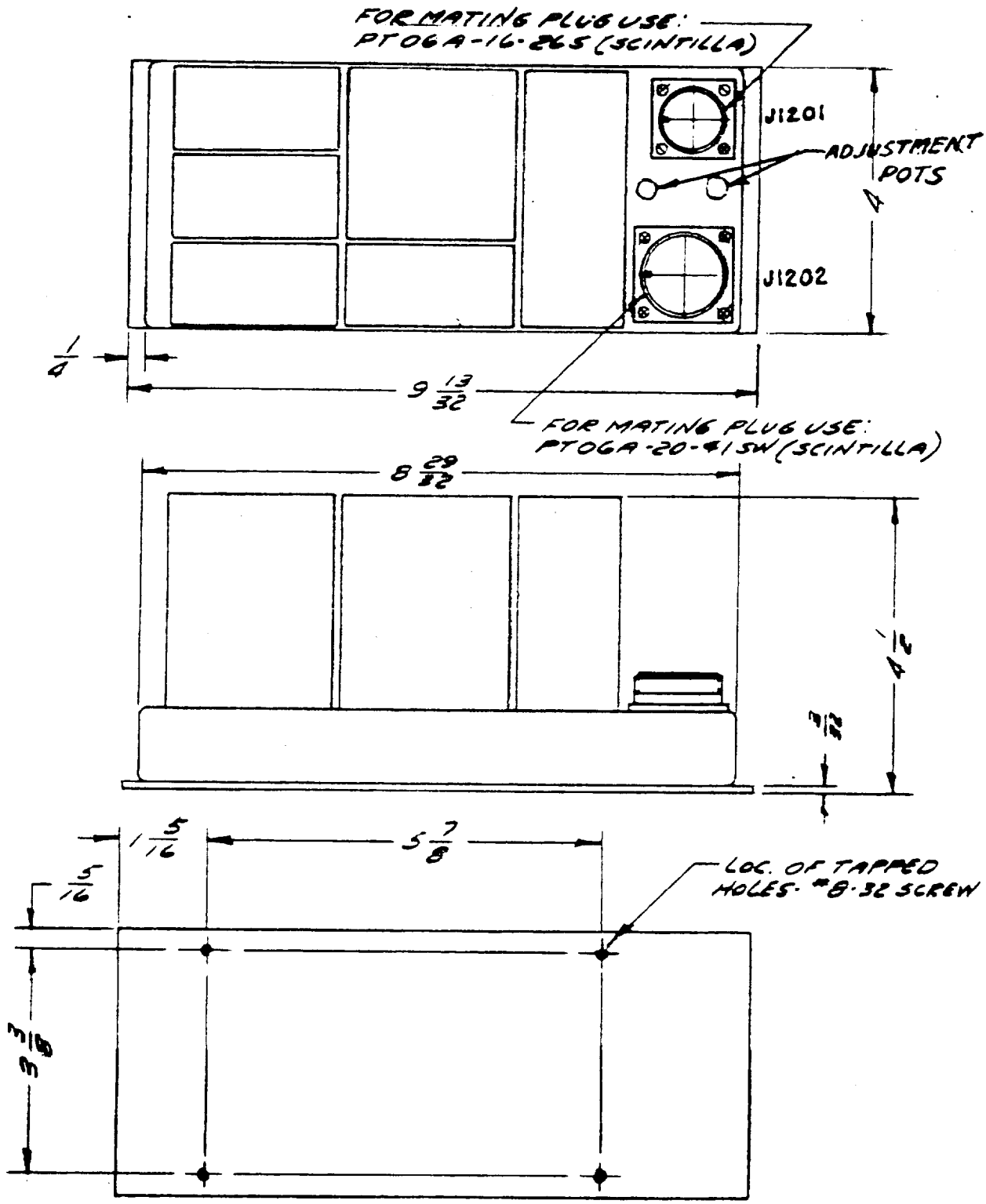


FIGURE 6. OUTLINE, AMPLIFIER, ALTITUDE, VERTICAL SPEED

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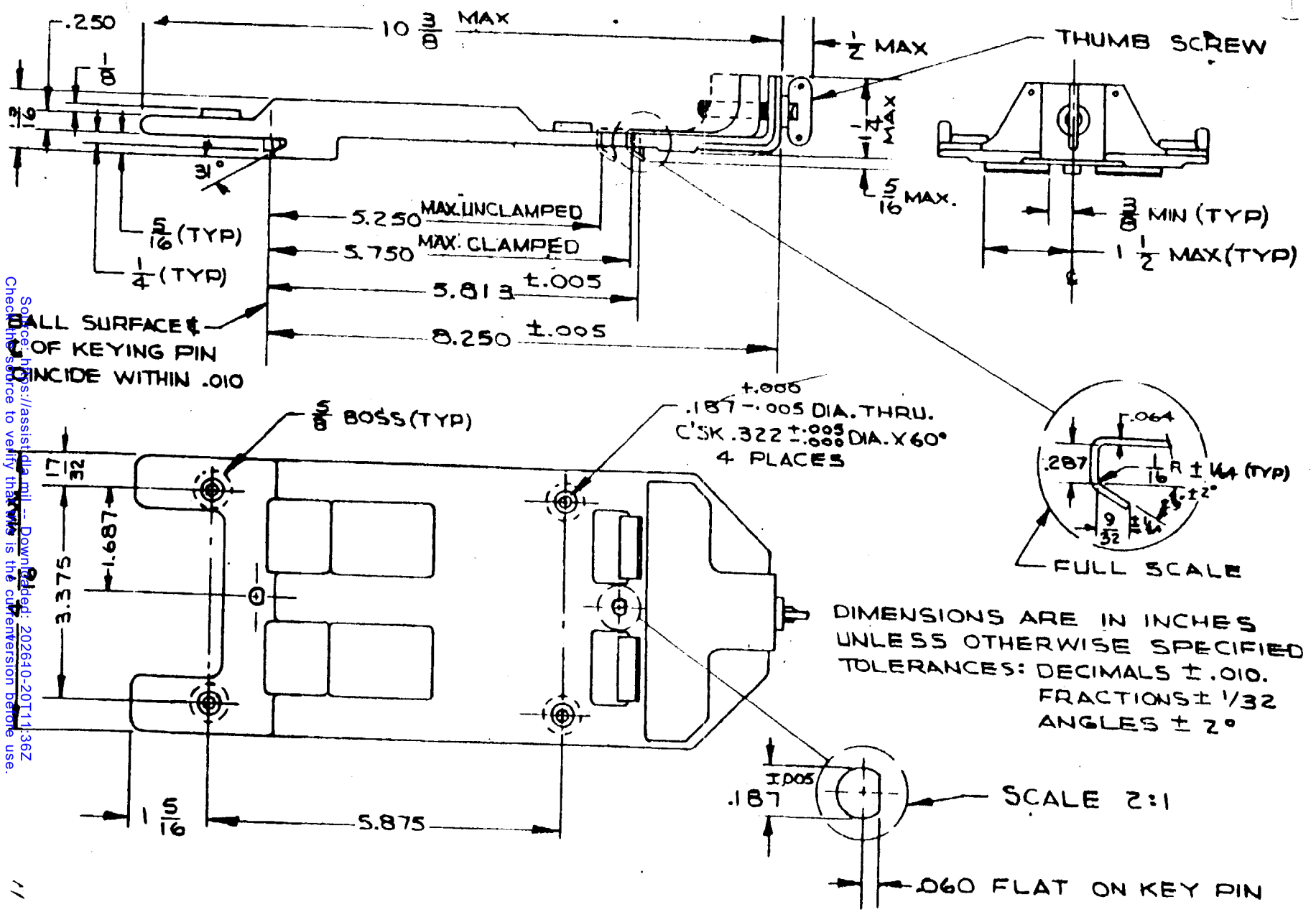


FIGURE 7. -MOUNTING BASE SCALE 1/2 = 1

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3.6.4 CONSTRUCTION.- The altitude-vertical speed display shall conform to figure 1. The design shall be reasonably simple to permit overhaul or repair without requiring numerous special tools and fixtures. The indicator shall be so constructed to withstand without failure, normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation and service.

3.7 PERFORMANCE.- The indicator and amplifier shall be capable of meeting the tests specified herein when subjected to the following conditions:

a. Vibration.- In accordance with MIL-E-5272, procedure IV and V for the indicator and procedure I for the amplifier.

b. Acceleration.- Procedure II of MIL-E-5272.

c. Humidity.- Relative humidity up to 100 percent.

d. Fungus.- Fungus growth as encountered in tropical climates.

e. Temperature.- -54°C to +71°C.

f. Sand and dust.- Sand and dust particles as encountered in de sert areas.

g. Temperature-altitude test.- The amplifier shall be capable of satisfactory operation under the various temperature-altitude conditions indicated on curves A, B, and C of figure 8.

h. Shock.- Procedure II of MIL-E-5272.

i. Altitude.- Pressures ranging from 31 to 0.820 inches Hg.

3.7.1 LIFE.- The systems shall be designed to have a reliable life of at least 1000 hours without requiring servicing. The system shall be designed to have a minimum operating life of at least 2000 hours with reasonable servicing and replacement of parts. The contractor shall prepare a list of parts requiring replacement within the minimum operating life of the equipment.

3.7.1.1 Life requirements of lamps shall be in accordance with applicable specifications approved by the procuring activity, and shall be replaced as required.

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SPECIFICATION CURVE FOR ALTITUDE V_s TEMPERATURE

a) CURVE A - DESIGN AND TEST REQUIREMENTS FOR CONTINUOUS OPERATION OF EQUIPMENT DESIGNED TO OPERATE AT 55 °C AT SEA LEVEL.

b) CURVE B - DESIGN AND TEST REQUIREMENTS FOR INTERMITTENT OPERATION OF EQUIPMENT DESIGNED TO OPERATE CONTINUOUSLY AT 55°C AT SEA LEVEL.

c) CURVE C - DESIGN AND TEST REQUIREMENTS FOR INTERMITTENT OPERATION OF EQUIPMENT FOR A MINIMUM PERIOD OF TEN MINUTES.

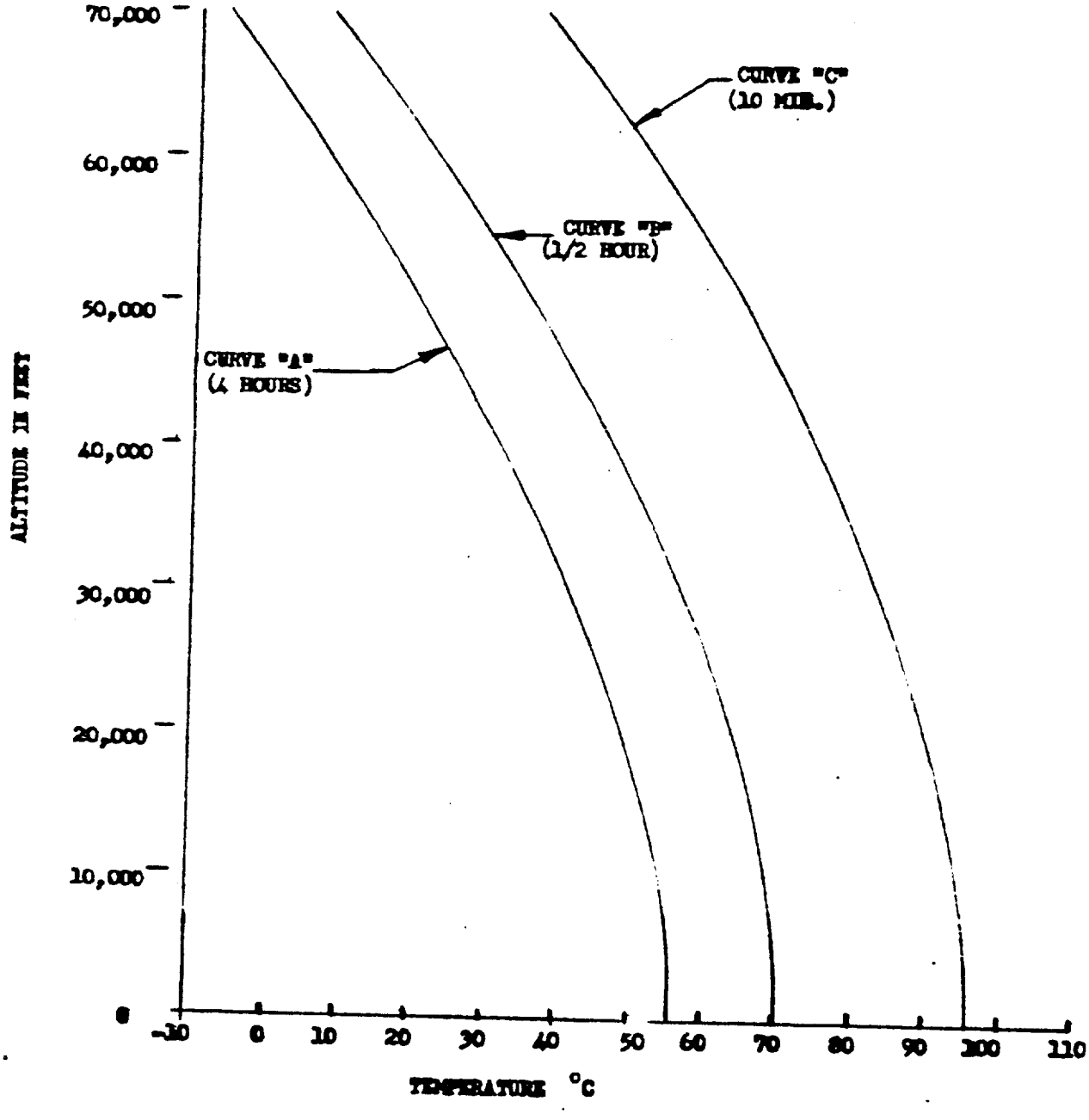


FIGURE 8. ALTITUDE VS TEMPERATURE

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3.7.2 RADIO INTERFERENCE.- The requirements of MIL-E-5400 shall apply except that the upper frequency limit of the radiated noise requirements shall be 1000 megacycles per second. The conducted noise requirements of MIL-L-26600 shall not apply to frequency range, nor shall they apply to those conductors which interconnect components.

3.8 INTERCHANGEABILITY.- All parts having the same manufacturers part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in part number shall be governed by the requirements of MIL-D-70327.

3.9 INDICATOR

3.9.1 CASE.- The indicator case shall be in accordance with figures 1 through 5. The case shall be nonferrous, low density material and shall be finished in durable dull black color No.37038 of FED-STD-595.

3.9.1.1 DIAL.- The dial layout and dimensions shall be in accordance with figures 1 and 2.

3.9.1.2 MARKINGS.- All markings shall be durable to withstand the usage encountered in service. The style and proportions of numerals and letters placed on the dial shall conform to figures 1 and 2.

3.9.1.3 NUMERALS.- Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be so placed that the center of mass of the numeral is on the line joining the appropriate graduations. Any confusion resulting in doubt as to which graduation the numeral applies, shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be approximately 1/64 inch.

3.9.1.4 DIAL TO LIGHTING WEDGE DISTANCE.- The distance from dial to lighting wedge shall be as small as practicable and shall not exceed 0.20 inch.

3.9.2 LIGHTING.- The indicator shall be internally lighted in accordance with MIL-I-25467 except number 337 lamps selected to 0.34 ±15 percent MSCP at 5 ±0.01V shall be used as the light source. Resistors may be used within the instrument to change the voltage applied to the lighting circuits. If resistors are used they shall be in series with the entire lighting circuit. Lighting acceptability is to be based on a visual comparison with a prime standard indicator only.

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3.9.2.1 PRIME STANDARD.- The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas covering the entire face of the indicator shall be measured for brightness level in fool-lamberts. The prime standard instrument and its recorded overall brightness measurements shall be approved by the procuring activity. The standard shall be checked periodically, at least every 100 hours of operation, to determine any change is brightness. Variations greater than ±20 percent in two or more areas shall require return of the standard for correction action.

3.9.3 INTEGRAL LIGHTING SUPPLY.- A 400 cps voltage, variable from 0 to 5V, rms shall supply power for internal lighting for both normal and emergency operation. The frequency tolerances of the supply voltage shall be as specified in MIL-E-5400.

3.9.4 METHOD OF PRESENTATION.- The method of presenting command and target altitude shall be by means of translational motion of symbols along vertical scales as well as indication on a counter. Sensitive altitude shall be presented by means of a moving scale against fixed index marks. Gross altitude shall be presented by means of a thermometer type band, indicating against a fixed altitude scale. Cabin pressure altitude shall be presented by means of a moving marker indicating against a fixed altitude scale. Vertical speed, between 0 and 2000 fpm ascent and descent, shall be presented by a moving pointer against a fixed scale. Rates in excess of 2000 fpm shall be presented by means of moving scales against a fixed index represented by the stalled pointer.

3.9.4.1 The indicator shall present the following information, derived from signal inputs in the form of synchro and rate generator signals from a central air data computer:

a. Aircraft altitude, sensitive, range -1000 to 80,000 feet.

b. Aircraft altitude, gross, range, -1000 to 80,000 feet.

c. Vertical speed, range ±40,000 fpm.

3.9.4.2 The following command information is derived from signal inputs from a remote coupler unit:

a. Command altitude, range 0 to 70,000 feet.

b. Target altitude, range 0 to 70,000 feet.

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3.9.5 BAROMETRIC SETTING OUTPUT.- The barometric, or zero setting system shall be mechanized as specified in 3.9.11 and the setting shall appear on a 4 digit counter and as the wiper position of a potentiometer having the following characteristics:

a. Resistance.- 20K ohms ±5 percent

b. Power rating.- 1.5w at 25°C

c. Rotation.- 350° ±1°

d. Resolution.- 1/800

e. Independent linearity ±0.5 percent.

3.9.6 AIRCRAFT ALTITUDE.- The aircraft altitude moving tape scales shall be actuated by a servo mechanism with a dual speed transmitting synchro input. The input to the instrument control transformer synchro shall be linearly proportional to altitude over the range of -1000 to 80,000 feet. Tape scale factor shall be 2 inches per 1000 feet.

3.9.6.1 AIRCRAFT ALTITUDE MECHANIZATION.- The servo drive mechanism shall be designed to accurately position the moving scale at the center reference line. Provision shall be made to prevent the moving scale from being driven beyond specified limits. Electrical zero of the follow-up synchro shall be at 29.92 inches Hg. Servo slew speed shall be not less than 80,000 fpm.

3.9.7 COMMAND ALTITUDE.- The command altitude symbols and associated counter shall be actuated by a servo mechanism with a dc voltage input proportional to command altitude.

3.9.7.1 COMMAND ALTITUDE MECHANIZATION.- The drive mechanism of the command altitude servo shall be designed to convert the rotary motion of the servo into translational motion of the symbol. The command altitude symbol shall be positioned to indicate zero altitude when the follow-up signal generator is in its electrical zero position. The mechanism shall be designed to accurately position the command marker with respect to the appropriate position on the aircraft altitude moving tape and command counter. When the command index is positioned to any value on the altitude scale, there shall be no apparent relative motion between command index and moving scale resulting from servoed motion of the moving scale in either direction. Motion of either servoed element shall not result in interaction with the other. When the command altitude is beyond the visible range of the instrument dial on the moving

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tape scale, provision shall be made to disengage the drive mechanism and the index shall assume a position at the scale extremity and remain visible at all times. The reading of the command altitude marker against the sensitive altitude tape (when the commanded value is within the visible range of the altitude scale) shall at all times read within 25 feet of the command value as read on the counter.

3.9.7.2 FOLLOW-UP POTENTIOMETER.- The follow-up potentiometer shall be of 50K ohm resistance ±5 percent. Power rating at 25°C shall be 2.5w. Independant linearity shall be ±0.1 percent. Rotation shall be 3600° +12° -0°. Servo slew speed shall be proportional to switch displacement from 0 to 80,000 fpm in manual mode and a maximum speed of not less than 80,000 fpm in automatic mode.

3.9.8 TARGET ALTITUDE.- The target altitude symbol and counter shall be actuated by a position servo mechanism with a dc voltage input proportional to target altitude.

3.9.8.1 TARGET ALTITUDE MECHANIZATION.- The mechanization shall be identical to that indicated in 3.9.7.1 and 3.9.7.2.

3.9.8.2 COUNTERS.- The target and command counters shall be of the continuous rotation type. Counter wheels which operate intermittently shall not be used.

3.9.9 SCALE FACTOR (COMMAND AND TARGET FUNCTIONS).- The scale factor for command and target functions shall be 0.001428V per foot. OV = 0 feet.

3.9.9.1 SCALE FACTOR (ALTITUDE INPUT FUNCTIONS).- The scale factor of the follow-up synchros shall be

36° per 1000 feet on the fine synchros
1.869° per 1000 feet on the coarse synchros

Bendix synchros AY-300, or equivalent, shall be used set to electrical zero at 29.92 inches Hg.

3.9.9.2 RANGE AND SCALE FACTOR - VERTICAL SPEED

AC Tachometer output 0.145V per 1000 fpm.
50 fpm to 40,000 fpm ascent and descent.

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3.9.10 SLEWING SWITCHES.- Manually operated slewing switches shall be provided as shown on figure 1 to operate the command symbols and associated counters. The switches shall be spring centered. The slewing speed imparted to the command symbol shall be proportional to the amount the switch is displaced from center within the minimum and maximum speeds specified. Up-position of the switch shall drive the symbol toward increasing value of the function. Operation of this switch will normally be under manual mode of flight during which time the remote servo signal input will be de-energized by the amplifier mode relay.

3.9.11 ZERO SETTING SYSTEM.- A zero setting system, controlled by a setting knob, shall be provided that will permit the altimeter to be set to indicate zero altitude at any existing ground level pressure throughout a range of 31 to 28 (inclusive) inches Hg. A safety feature shall be provided on the pressure scale whereby the scale cannot be read erroneously when the zero setting mechanism is moved beyond the range of 31 to 28 inches Hg. The pressure scale shall be a four digit counter as shown on figure 1. Clockwise rotation of the knob shall drive the counter numerals toward increasing numerical value.

3.9.11.1 ADJUSTMENT OF SETTING.- The zero setting system shall be provided with a suitable means to adjust or correct the relative setting of the altitude and the zero setting system indications, should this become necessary. The adjustment shall be accessible from the face of the altimeter when installed on the instrument board. Range of adjustment shall be ±200 feet. A protective covering shall be provided to prevent tampering with the adjustment.

3.9.12 POWER FAILURE INDICATOR.- In the event of normal and emergency power failure, a spring loaded drop out flag, red in color, with black OFF lettering and having automatic reset when power is restored, shall be provided. One flag shall be provided for the sensitive altitude scale. Location and configuration shall be as specified on figure 1.

3.9.13 CABIN PRESSURE INLET ORIFICE.- A cabin pressure orifice having a diameter of 1/4 inch ±1/16 inch shall be provided on the rear side of the case as shown on figure 3. The orifice shall be protected by a 150 mesh corrosion proof screen.

3.10 AMPLIFIER.- The servo amplifier employed shall be a transistor or transistor-mag amplifier. It shall be modular in construction with plug-in modules for the power supply, auxiliary circuits and amplifier circuits. Envelope dimensions shall not exceed those specified on figure 6, excluding the shockmount. The mounting base for the amplifier shall be in accordance with figure 7.

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3.10.1 INPUT IMPEDANCE.- Input impedance for the command and target altitude servos shall be such that the maximum current drain shall not exceed 1 ma per function.

3.10.2 DC POWER LOADING.- The command functions power requirements shall not exceed 1 ma per signal channel for 100V dc input.

3.10.3 MODE RELAYS.- The command functions shall be switched independently by separate mode relays in the amplifier. The relays will be de-energized for "manual mode". Upon input of 28V dc signal to the relay system the mode relays will accomplish all necessary transfer functions to change the mode of operation from manual to automatic.

3.10.4 SERVO RESPONSE.- Response of target, command and altitude servos shall be as listed below:

Function	Bandwidth	Damping Ratio
Target altitude	2 - 3	0.4 to 0.7
Command altitude	2 - 3	0.3 to 0.7
Altitude	3 - 6	0.2 to 0.7

3.10.4.1 RESPONSE OF VERTICAL SPEED DISPLAY.- The vertical speed servo loop and associated display shall respond in an exponential manner to a step function input of vertical speed of 2000 fpm. The time constant of this response shall have a minimum range from 2 to 6 seconds. Methods shall be provided on the amplifier to accomplish this adjustment. Bandwidth shall have a minimum range of 0.04 to 0.4 cps and damping shall be critical or greater.

3.10.5 ZERO SET ADJUSTMENT.- An adjustment shall be provided on the amplifier to permit adjustment of the vertical speed index to the zero graduation. Range of this adjustment shall be ±200 fpm.

3.10.6 FEED BACK POTENTIOMETER.- The follow-up potentiometer for vertical speed shall be a shaped function pot with 20K resistance tapped and shunted to give nonlinear response in accordance with nonlinear tape scale factor as follows:

a. From 2000 to 6000 fpm, 1000 fpm = 1/2 inch

b. From 6000 to 10,000 fpm, 2000 fpm = 1/2 inch

c. From 10,000 to 40,000 fpm, 5000 fpm = 1/2 inch

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3.10.7 EXCITATION.- Excitation for all remote transmitting synchros or potentiometers shall be furnished from the indicator amplifier unless otherwise specified.

3.10.7.1 Reference voltage for command target feed back potentiometers shall be supplied by the command data source equipment.

3.10.8 ELECTRON TUBES.- Electron tubes shall not be used in the equipment covered by this specification.

3.11 CONNECTORS.- Electrical connectors used on the indicator and amplifier case shall be of hermetic seal type which electrically conform to MIL-C-5015. Miniature connectors, Scintilla Type, as specified herein, or the electrical equivalent shall be utilized. Keying provisions shall be made where necessary. Connector pin arrangements shall be as specified in table I.

3.11.1 EXTRA CONTACTS IN RECEPTACLES.- The requirement for extra contacts shall not apply unless it is deemed necessary for conducting tests.

3.12 SCREW THREADS.- Unless otherwise specified, the threads of all machine screws, 0.060 of an inch or larger in diameter, shall conform to MIL-S-7742.

3.13 WEIGHT.- The weight of the indicator system shall be held to a minimum consistent with design requirements. The indicator weight shall not exceed 9 pounds 14 ounces, the amplifier shall not exceed 8 pounds including the mounting base.

3.14 FINISHES AND PROTECTIVE COATINGS

3.14.1 PLATING.- Plating is not required on normally lubricated parts, such as gears, bearings and shafts, fabricated from brass, bronze or corrosion resistant steel.

3.14.2 CADMIUM PLATED PARTS.- Cadmium plated parts shall be in accordance with QQ-P-416, of a type and class to be determined by their application.

3.14.3 CASE.- The case shall be of nonferrous, low density material and shall be finished in durable dull black.

3.14.4 ALUMINUM ALLOY PARTS.- Aluminum alloy parts shall be covered with an anodic film in accordance with MIL-A-8625 except that small holes, straight threads and case inserts need not be anodized.

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TABLE I

Pin Connection Table For Altitude, Vertical Speed Indicator

Connector PT02A-20-41PW PT06A-20-41SW Pin-Amplifier	to	Connector PT06A-20-41SW PT1E-20-41PW Pin-Indicator	Function	Section
A		A	Pot excitation	Vert. speed
B		B	Fine autosyn output H_2 and rate generator	Altitude
C		C	Rate generator excitation	Altitude
D		D	Rate generator excitation	Command altitude
E		E	Signal ground	Target altitude
F		F	Power warning flag (-)	
G		G	Spare	
H		H	Spare	
J		J	Spare	
K		K	Chassis ground	
L		L	Amplifier output	Target altitude
M		M	Chopper input-Pot wiper	Command altitude
N		N	Rate generator output	Command altitude
P		P	Rate generator output	Altitude
R		R	Pot excitation Lo	Vertical speed
S		S	Power ground	Vertical speed
T		T	Amplifier output	Vertical speed
U		U	Rate generator output	Vertical speed
V		V	rate generator output	Vertical speed
W		W	Capacitor	Vertical speed
X		X	Coarse autosyn output H_2	Altitude
Y		Y	Motor voltage fixed ϕ	Altitude
Z		Z	Motor voltage fixed ϕ	Command altitude
a		a	Chopper input-pot wiper	Target altitude
b		b	Power warning flag (+)	
c		c	Motor voltage fixed ϕ	Target altitude
d		d	Capacitor	Target altitude
e		e	Rate generator excitation	Target altitude
f		f	Rate generator output	Command altitude
g		g	Fine and Coarse autosyn to signal ground C_2 and C_2	Altitude
h		h	Pot wiper	Vertical speed
i		i	Rate generator excitation	Vertical speed
j		j	Motor voltage fixed ϕ	Vertical speed
k		k	Capacitor	Altitude
m		m	Capacitor	Command altitude
n		n	Rate generator output	Target altitude
p		p	rate generator output	Target altitude

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TABLE I (Con't)

q	q	Signal ground	Command altitude
r	r	Amplifier output	Altitude
s	s	Signal ground	Vertical speed
t	t	Amplifier output	Command altitude

Connector PT02A-16-26PW PT06A-16-26SW Pin Amplifier	to	Connector PT06A-14-19S PT1H-14-19P Pin Indicator	Function	Section
A			Tachometer excitation Hi	Vertical speed
B			Tachometer output Hi	Vertical speed
C			115V 400 cps	
D			Spare	
E			Pot wiper from indicator 2 (dual installation)	Command altitude
F			Spare	
G			Spare	
H			Spare	
J			Spare	
K			Spare	
L			Signal ground	Command altitude
M		R	Motor fixed ϕ Lo	Command altitude
N			Rotor leads C_2 power ground	Altitude
P			Signal ground	Vertical speed
R			Tachometer excitation Lo	Vertical speed
S			115V 400 cps	
T			Rotor leads H_2 transmitter excitation	Altitude
U			Chopper input	Command altitude
V			28V dc relay voltage	Command altitude
W			Chopper input	Target altitude
X			Spare	
Y			Rate divider (for command switch dual installation)	Command altitude
Z			28V dc relay voltage	Command altitude
a			Chopper output-dual installation	Command altitude
b		V	Slew input - Slew pot output	Command altitude
c			Motor voltage ϕ, Lo (Dual installation) for Command switch	

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TABLE I (Con't)

Connector PT1H-14-19P PT06A-14-19S Pin-Indicator	to	Connector PT06A-16-26SW PT02A-16-26PW Pin-Amplifier	Function	Section
A			Lights 5V	
B			Lights 5V	
C			Coarse autosyn stator lead X2	Altitude
D			Motor Voltage fixed ϕ Lo (dual installation)	Command Altitude
E			Barometric setting - Excitation Lo	
F			Spare	
G			Pot wiper (dual instal)	Command altitude
H			Barometric setting - wiper	
J			Pot excitation +100V dc	
K			Coarse autosyn stator lead Y2	Altitude
L			Fine autosyn stator lead X2	Altitude
M			Tachometer output Lo to pot center tap	Vertical speed
N			Fine autosyn stator lead Y2	Altitude
P			Fine autosyn stator lead Z2	Altitude
R		M	Motor fixed ϕ Lo	
S			Barometric setting - excitation Hi	
T			Pot excitation ov	
U			Coarse autosyn stator lead Z2	
V		b	Slew pot output	

3.14.5 COVER GLASS AND LIGHTING WEDGE.- All reflecting surfaces shall be coated with magnesium flouride.

3.15 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.16 WORKMANSHIP

3.16.1 GENERAL.- The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.16.2 DIMENSIONS.- Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the altimeter, they shall be held or limited accordingly.

3.16.3 GEARS.- Gear assemb lies shall be properly aligned and meshed, and shall be operable without interference, tight spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.16.4 CLEANING.- The altimeter shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- Inspection and testing of the indicator and amplifier shall be classified as follows:

a. Acceptance tests.- See 4.2

b. Preproduction tests.- See 4.5

4.2 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests

4.2.1 INDIVIDUAL TESTS.- Each indicator and amplifier shall be subjected to the following tests as described under TEST METHODS 4.4:

4.2.1.1 INDICATOR

a. Examination of product

b. Functional check

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c. Synchro correspondence

d. Case leak

e. Dielectric

f. Lighting

g. Scale error

h. Servo slew speed

i. Position error

j. Power warning flag

k. Friction

4.2.1.2 AMPLIFIER

a. Examination of product

b. Leakage test

c. Functional test

d. Insulation breakdown

e. Insulation resistance

f. Continuity tests

g. Relay test

h. Voltage and current tests

i. Power consumption

j. Signal channel measurements

4.2.2 SAMPLING PLANS AND TESTS

4.2.2.1 SAMPLING PLAN A.- One amplifier and one indicator selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.4 TEST METHODS.

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4.2.2.1.1 INDICATOR

a. Individual tests

b. Scale error at -55°C

c. Scale error at 71°C

d. Vibration error, procedure IV

e. Frequency response

f. Sampling plan A of MIL-L-25467

4.2.2.1.2 AMPLIFIER

a. Low temperature operation

b. High temperature operation

c. Vibration

4.2.2.2 SAMPLING PLAN B.- Unless otherwise specified, 3 indicators and 3 amplifiers selected at random from the first 15 items on contract shall be subjected to the following tests as described under 4.4 TEST METHODS.

4.2.2.2.1 INDICATOR

a. Sampling plan A

b. Magnetic property

c. Vibration failure

d. High altitude

e. Acceleration

f. Humidity

g. Fungus

h. Salt spray

i. Endurance

j. Sampling plan B of MIL-L-25467

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4.2.2.2.2 AMPLIFIER

a. Temperature exposure

b. Temperature-altitude tests

c. Shock

d. Acceleration

e. Humidity

f. Radio interference

g. Salt spray

h. Fungus

i. Magnetic property

j. Endurance

4.2.2.3 REJECTION AND RETEST.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.2.2.3.1 INDIVIDUAL TESTS MAY CONTINUE.- For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.2.3 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS

4.3.1 INDICATOR

4.3.1.1 STANDARD ATMOSPHERIC CONDITIONS.- Whenever the pressure and temperature are not specified, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are

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made with atmospheric pressure or temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

4.3.1.2 VIBRATION.- Unless otherwise specified, the indicator shall not be subjected to any form of vibration or tapping during the performance of any of the tests specified in the specification.

4.3.1.3 ATTITUDE.- Unless otherwise specified, the indicator shall be tested in the normal operating position.

4.3.1.4 ELECTRICAL POWER.- Unless otherwise specified, the indicator shall be tested with 115V 400 cps ±2 percent single phase ac power, 100V ±2 percent dc power, and 28V ±2 percent power.

4.3.1.5 STANDARD TEST EQUIPMENT.- The indicator shall be tested with standard AY-201S-1-B synchros (or equivalent) and precision potentiometers, resistance 10K ohms ±5 percent; 2.5w at 25°C, resolution 0.01 percent.

4.3.1.6 TEST SYNCHROS.- The synchro transmitters used to provide indicator input functions shall be accurate to ±0.1° either by selection or calibration.

4.3.1.7 TEST POTENTIOMETERS.- The potentiometers used to provide indicator input functions shall have a linearity of at least 0.005 percent.

4.3.1.8 STANDARD TEST AMPLIFIER.- The standard test amplifier shall be a standard production amplifier which has successfully passed all individual tests.

4.3.1.9 ELECTRICAL ZERO AND ROTATION REFERENCE.- The indicator synchros shall be so oriented electrically that the altitude function shall be servoed to the indication listed in the table below when a standard test transmitter is set at electrical zero and connected to the indicator and amplifier as per the table below. The pins of amplifier connector J1202 shall be connected to the identically lettered pins of indicator connector J1102. The altitude coarse synchro shall be checked simultaneously with the fine synchro. After the function is servoed to the EZ point by the fine autosyn, the coarse synchros shall be excited as per figure 9. The vacuum tube voltmeter must indicate less than 0.150V. A clockwise rotation of the standard test transmitter rotor (viewed from the shaft end) shall cause increasing function. A standard test transmitter is defined as follows: The rotor leads are arbitrarily labeled H_2 and C_2. One stator lead is arbitrarily

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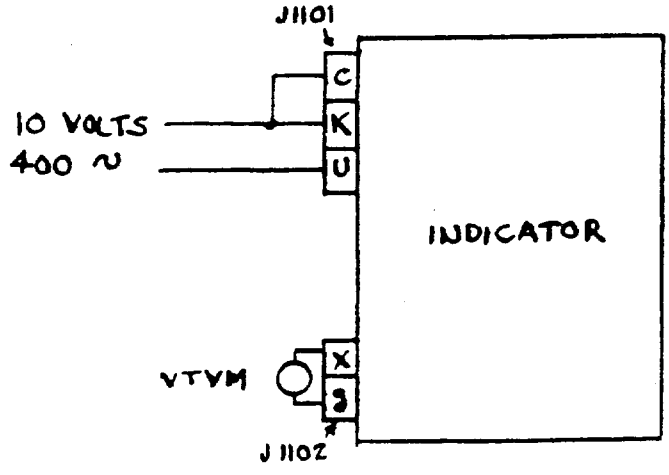


FIGURE 9. ALTITUDE COURSE SYNCHRO

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labeled Z_2. Z_2 is connected to C_2 and 26V 400 cps is applied across H_2 - C_2. At electrical zero there will be a null voltage measured across the two unlabeled stator leads (V_1). The voltage between H_2 and either of the unlabeled stator leads shall be 15V (approximate) (V_2). If this voltage is 38V (approximate), the synchro is 180° away from KZ. See figure 10. If cw (clockwise) rotation (viewed from the shaft end) of the synchro rotor causes V_3 to increase (within 30° of KZ) V_3 is between X_2 and Z_2, if the voltage decreases V_3 is between Y_2 and Z_2.

Electrical Zero Test

Function	Indicator reading with transmitter set at KZ	Standard test transmitter pin connections			
		H_2	C_2	X_2	Y_2	Z_2
Altitude	0 feet	T/J1201	N/J1201	L/J1101	N/J1101	P/J1101

4.3.1.10 POTENTIOMETER ZERO.- Unless otherwise specified, the low end of the potentiometer will correspond to the low end of the function.

4.3.2 AMPLIFIER

4.3.2.1 TEST EQUIPMENT.- The test equipment shall not load or affect the normal operation of the amplifier. The tolerances specified are acceptable limits of performance and are not to be confused with meter accuracies. The following instruments or their equivalents may be used:

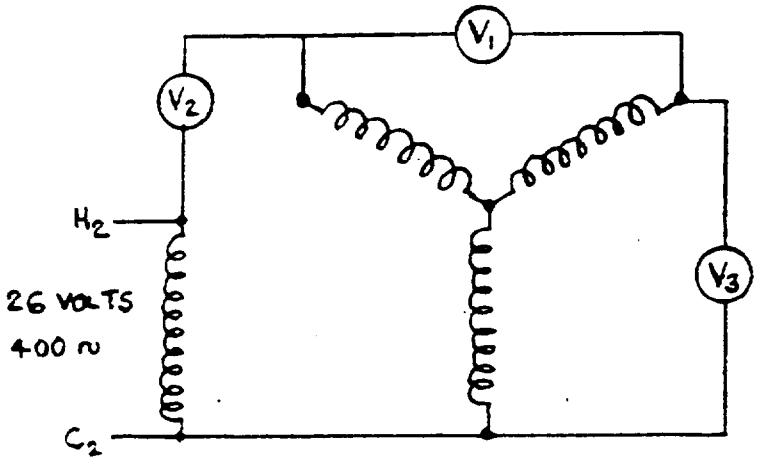
a. Vacuum Tube Voltmeter Model 400C
Hewlett-Packard Co.
Palo Alto, California

b. Megger, James Biddle Co.
Philadelphia, Pennsylvania

c. Cathode-Ray Oscilloscope, Type 304A
Allen B. Dumont Labs, Inc.
Clifton, New Jersey

d. Voltohmmeter-Senior Volt Ohmyst WV-98A
Radio Corporation
Harrison, New Jersey

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V_1 IS A VACUUM TUBE VOLTMETER
V_2 IS 1000Ω/VOLT VOLTMETER
V_3 IS 1000Ω/VOLT VOLTMETER

FIGURE 10. ELECTRICAL ZERO CIRCUIT

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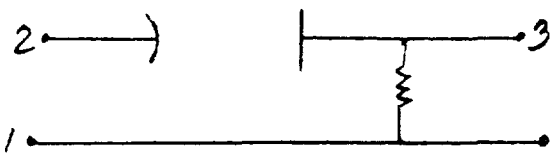
e. Phase Comparator, TMS 113-4-A
Eclipse-Pioneer
Teterboro, New Jersey

4.3.2.2 TEST POSITION.- Unless otherwise specified the amplifier shall be tested in its normal operating position.

4.3.2.3 POWER SUPPLY

4.3.2.3.1 ALTERNATING CURRENT.- Where normal excitation is specified, it shall be 115V ±3V rms, 400 cps single phase source with a total harmonic distortion less than 5 percent and a frequency deviation less than 1 percent capable of supplying 150 va minimum.

4.3.2.3.2 PHASE.- A voltage specified as being "from pin X to pin Y" shall be measured using pin X as the reference or ground terminal. The direction of leading phase angle can be established by use of the following circuit:



In this circuit the voltage from pin 1 to pin 3, V_{13}, leads the voltage from pin 1 to pin 2, V_{12}. All phase measurements shall be taken with respect to the ac excitation measured from pin C to pin S, V_{CS}, on connector J1201.

4.3.2.3.3 DIRECT CURRENT POWER SUPPLY.- The 28V ±1V dc supply shall be capable of supplying 0.150 amp. Ripple and noise on the dc voltage not to exceed 5 percent.

4.3.2.4 AC SIGNAL SOURCE.- The 400 cps source shall have the following characteristics:

a. Output impedance.- Not to exceed 1000 ohms.

b. Voltage range.- 0 to 1V continuously variable.

c. Phase.- 0° and 180° ±3° selectable.

d. Distortion.- Less than 5 percent.

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4.3.2.5 LOAD.- The load (Z) specified herein shall be 155)40° ohms ±1 percent, measured at 26V, 400 cps.

4.3.2.6 DC SIGNAL SOURCE.- The dc signal source shall have the following characteristics:

a. Output resistance.- The output resistance shall be less than 1000 ohms.

b. Voltage range.- 0 to 1V continuously variable.

c. Polarity.- Both polarities shall be available (that is positive and negative).

d. Ripple.- The total rms voltage appearing at the signal source shall not exceed 2 mv.

4.4 TEST METHODS

4.4.1 INDICATOR TESTS

4.4.1.1 EXAMINATION OF PRODUCT.- Each indicator shall be inspected to verify that all parts have been constructed in a workmanlike manner. Attention shall be given to neatness, thoroughness of soldering, wiring, marking of parts and assemblies and freedom of parts from burrs and sharp edges.

4.4.1.2 FUNCTIONAL CHECK.- The following functional check shall be performed only on the completed indicator. The indicator functions shall be simultaneously cycled at the rate of one cycle per 2 minutes for 2 hours. A cycle shall consist of the excursion from minimum value of the function, to maximum value, and return to minimum value. This test to be performed with the system in the automatic mode. The indicators shall meet the requirements of this specification after completion of the functional check.

4.4.1.3 SYNCHRO CORRESPONDENCE.- Any indicator section that has a dual speed input shall have these synchros inspected for assurance that the coarse synchro is set in satisfactory correspondence with its associated fine synchro. The coarse synchro shall be considered in satisfactory correspondence when the rotor output is less than 150 mv with the stator excited as in figure 9 for a fine synchro setting of precisely EZ. The 10V supply may be replaced by a precision synchro transmitter set precisely at EZ.

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4.4.1.4 CASE LEAK.- Each indicator shall be filled with a 10 percent helium, 90 percent nitrogen mixture at a pressure of 1/2 atmosphere, dust free and with a dew point below -65°C. Each shall be tested for case leaks following the final sealing. The leak rate shall not exceed 7.5 micron cubic foot per hour.

4.4.1.5 DIELECTRIC STRENGTH TEST.- The application of 250V 60 cps for a 5 second interval between the following pins and the shell of the connector shall cause no evidence of voltage breakdown:

J1102 F, M, R, J, g, n, q

J1101 A, H, U, V, L

4.4.1.6 LIGHTING (VISUAL COMPARISON).- Indicator lighting shall be checked by comparison with a prime standard. The prime standard and the instrument under test shall be placed side by side, and at eye level. Both units shall be connected to a common variable source (0-5V). The overall lighting of the two units shall be compared with 2.85V ±0.1V applied to the standard and 2.85V ±0.1V applied to the unit under test. If the general overall brightness appears the same visually, the unit under test shall be considered acceptable. If any area of the instrument under test appears visually to differ from the same area of the prime standard an additional test shall be made as follows: The common voltage source shall be gradually reduced. The area in question of the indicators under test and the comparable area of the prime standard shall extinquish at the same voltage within ±5 percent.

4.4.1.6.1 STRAY LIGHT.- The lighting system shall be so housed as to prevent the leakage of stray light and to shield from direct view all lamp filaments. Stray lightmeasurements shall be made as follows: A flat white reflecting surface of 85 ±5 percent reflectance shall be placed perpendicular to the front cover glass and parallel to the top edge of the mounting flange. Brightness readings shall be made looking down (perpendicular) on this reflecting surface and 1 inch in front of the cover glass. When this reflecting surface is positioned in the lower half of the display the readings shall not exceed 1.5 foot lamberts, and when positioned in the upper half shall not exceed 0.2 foot-lamberts. Measurements shall be repeated looking upon the reflecting surface and in no case will the readings exceed 0.2 foot-lamberts.

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4.4.1.7 SCALE ERROR

4.4.1.7.1 SENSITIVE ALTITUDE.- The coarse synchro shall be set at the value specified in table II. The tape shall be brought into the specified reading using the fine synchro only, being careful to approach from the right direction. The barometric counter shall be set at 29.92. The tape may be initially adjusted tozero altitude for dual synchro settings of zero altitude by use of the screwdriver slot located below the barometer set knob. With both synchros set at EZ, the barometer set knob set at 29.92 and the altitude tape adjusted to zero indication, the tape adjustment screw shall permit an adjustment range of ±200 feet ± 30 feet.

TABLE II

Sensitive Altitude

Reading Altitude Feet	Transmitter Coarse	Angle Fine
0	0	0
1,000	1.9°	36°
2,000	3.7°	72°
4,000	7.5°	144°
5,000	9.3°	180°
10,000	18.7°	0
20,000	37.4°	0
30,000	56.1°	0
40,000	74.8°	0
50,000	93.5°	0
60,000	112.1°	0
70,000	130.8°	0
80,000	149.6°	0

4.4.1.7.1.1 ACCURACY.- Test readings shall be obtained for both function increasing and function decreasing. All readings shallnot be in excess of 0.9° on the fine synchro transmitter (25 feet). The gross tape shall clearly indicate the correct altitude graduation.

4.4.1.7.1.2 SENSITIVITY.- Rotation of the fine synchro transmitter ±0.36° (10 feet altitude) shall result in apparent motion of the sensitive altitude tape.

4.4.1.7.2 COMMAND ALTITUDE (AUTOMATIC MODE).- This test may be run concurrently with sensitive altitude. After the tape has been positioned at the lubber line, the command altitude marker shall be positioned directly under the lubber line, being careful to approach from the right direction. The transfer function shall

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be read off the transmitter. The counter shall then be positioned at the correct altitude value, being careful to approach from the right direction. The transfer function shall be read off the transmitter.

4.4.1.7.2.1 ACCURACY.- The transfer function readings taken in 4.4.1.7.2 shall not differ from the values set in table III by more than 0.0014. The gross marker shall clearly indicate against the correct altitude graduation at all times.

TABLE III

Command Altitude

Reading Feet	Voltage Output and Voltage Input
0	0
2,000	0.0286
4,000	0.0571
10,000	0.1429
20,000	0.2857
30,000	0.4286
40,000	0.5714
50,000	0.7143
60,000	0.8572
70,000	1.0000

4.4.1.7.2.2 CORRELATION.- Not only must each presentation meet its individual accuracy requirements but they must also maintain close correlation between themselves.

4.4.1.7.2.3 CORRELATION ACCURACY.- The marker transfer function shall not differ from the counter transfer function at any station by more than 0.0004 (25 feet).

4.4.1.7.2.4 DISENGAGEMENT.- When the transfer function commands a position beyond the viewing range of the indicator, the marker shall position itself at the window extremities, in view, and resume indexing against the altitude tape when that portion of the tape is stationed in the window area.

4.4.1.7.2.5 DISENGAGEMENT TEST.- With dual speed synchros set in 0 altitude on the tape. Set in 70,000 feet on the potentiometer command transmitter. The marker shall position itself at the upper extremity of the window. Set in 70,000 feet with the dual synchro inputs. The tape shall position the marker such that some part of

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the marker is obscured by the lubber line. Set in 0 feet on the potentiometer command transmitter. Set in 0 feet with the dual synchro inputs. The tape shall position the marker such that some part of the marker is obscured by the lubber line. The counter and gross marker will display the command altitude throughout.

4.4.1.7.3 MANUAL MODE.- The manually operated slew switch below the altitude window actuates the command marker and command counter. Up position of the switch shall increase command function. Down position shall decrease command function. The rate of slew shall be proportional to switch displacement.

4.4.1.7.4 BAROMETRIC CORRECTION OPERATIONAL CHECK.- With the sensitive altitude tape set at exactly +1000 feet, the barometric correction mechanism shall be checked at the stations specified in table IV, for both increasing and decreasing function.

TABLE IV

Barometric Correction

Barometric Pressure	Altitude Feet
31.00	1983
30.73	1740
30.51	1540
30.28	1330
29.92	1000
29.63	730
29.48	590
29.29	410
28.98	120
28.75	-100
28.50	-340
28.00	-824

4.4.1.7.4.1 ACCURACY.- All readings shall not be in error in excess of 15 feet.

4.4.1.7.5 SCALE ERROR FOR SEA LEVEL PRESSURE OUTPUT POTENTIOMETER. The sea level pressure potentiometer output voltage shall be directly proportional to the barometric pressure as read on the counter as specified in table V, in both increasing and decreasing directions.

4.4.1.7.5.1 ACCURACY.- All readings shall not be in error in excess of 0.015 inches Hg.

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TABLE V

Sea Level Pressure Potentiometer

Barometric Pressure	Voltage Output and Voltage Input
31.00	1.000
30.73	0.9100
30.51	0.8367
30.28	0.7600
29.92	0.6400
29.63	0.5433
29.48	0.4933
29.29	0.4300
28.98	0.3267
28.75	0.2500
28.50	0.1667
28.00	0

4.4.1.7.6. ALTITUDE ZERO ADJUSTMENT.- Rotation of the zero set knob, located just below the barometric knob, from one extreme to the other shall result in a movement of ±200 ±30 feet on the sensitive altitude tape, within the limits of -1000 to +80,000 feet as read on the tape.

4.4.1.7.7 SCALE ERROR TEST FOR VERTICAL SPEED.- Vertical speed between 0 and 2000 fpm ascent and descent, is presented by the moving pointer against the fixed scale. Rates in excess of 2000 fpm are presented by the moving tape as read against the stalled pointer. Tests shall be conducted at points specified in table VI.

TABLE VI

Vertical Speed

Reading Feet Per Minute	Tolerance Feet Per Minute	Voltage Output and Voltage Input	Tolerance
40,000	±500	1.000	0.0125
25,000	±500	0.625	0.0125
8,000	±190	0.200	0.0047
4,000	±100	0.100	0.0025
2,000	±100	0.050	0.0025
1,000	±75	0.0250	0.0019
0	±75	0	0.0019
-1,000	±75	0.0250	0.0019
-2,000	±100	0.050	0.0025

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TABLE VI (Con't)

Reading Feet Per Minute	Tolerance Feet Per Minute	Voltage Output and Voltage Input	Tolerance
-4,000	±100	0.100	0.0025
-8,000	±190	0.200	0.0047
-25,000	±500	0.625	0.0125
-40,000	±500	1.000	0.0125

4.4.1.7.8 TARGET ALTITUDE.- With the input transfer function as indicated in table VII, the target counter and marker shall be read against the gross scale, in both increasing and decreasing direction.

TABLE VII

Target Altitude

Reading Feet	Voltage Output and Voltage Input
0	0
2,000	0.0286
4,000	0.0571
6,000	0.0857
8,000	0.1143
10,000	0.1429
15,000	0.2143
20,000	0.2857
30,000	0.4286
35,000	0.5000
40,000	0.5714
50,000	0.7143
60,000	0.8572
70,000	1.0000

4.4.1.7.8.1 ACCURACY.- All counter readings shall not be in error in excess of 300 feet. The marker shall clearly indicate against the correct altitude graduation at all times.

4.4.1.7.9 SCALE ERROR TEST FOR CABIN PRESSURE.- Cabin pressure accuracy, as read against the gross scale, shall be as specified in table VIII. Light tapping of the indicator case shall be permitted during this test.

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TABLE VIII

Cabin Pressure

Input Pressure Inches Hg.	Reading Feet	Tolerance
29.92	0	±500
25.84	4,000	±500
22.22	8,000	±500
19.03	12,000	±500
16.21	16,000	±500
13.75	20,000	±600
11.59	24,000	±700
9.72	28,000	±800
8.10	32,000	±900
6.71	36,000	±1000
5.54	40,000	±1100
4.36	45,000	±1200
3.44	50,000	±1300

4.4.1.8 SLEW SPEED TEST.- Using a stop watch, record the time for full excursion of the various presentations listed in table IX. The times shall not exceed those specified for either increasing or decreasing direction.

TABLE IX

Slew Speed

Function	Maximum Time In Seconds
Sensitive altitude tape, 0-80,000	60
Command altitude, 0-70,000	50
Target altitude 0-70,000	50
Vertical speed, -40,000 to +40,000 fpm	20
Command altitude (Manual) 0-70,000 (Max. switch displacement-room test only)	50

4.4.1.9 POSITION ERROR.- With all servoes at null, in the normal operating position, record the readings. These readings shall not vary with the indicator held in any other position.

4.4.1.10 POWER-OFF WARNING FLAG.- When the indicator is connected to the test amplifier and properly excited, the warning flag shall disappear from view.

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4.4.1.11 FRICTION TEST.- The indicator shall be operated throughout its entire range and the motion of the tapes, command bars, and indices carefully observed. There shall be no evidence of tape stickage or erratic motion of any kind of the moving components.

4.4.1.12 SCALE ERROR AT -55°C.- The indicator shall be subjected to a temperature of -55°C ±2°C for a period of 4 hours with no power applied. At the end of this period and after a 3 minute warm-up, while still at the specified temperature, the indicator shall meet the requirements of 4.4.1.7.1, 4.4.1.7.1.1, 4.4.1.7.2, 4.4.1.7.7, 4.4.1.7.8, 4.4.1.7.9, 4.4.1.8 and 4.4.1.10. The permissible tolerances shall be increased by 75 percent.

4.4.1.13 SCALE ERROR AT +71°C.- The indicator shall be subjected to a temperature of +71° ±2°C for a period of 4 hours with no electrical power applied. At the end of this period and while still at the specified temperature the indicator shall meet the requirements of 4.4.1.12 except that permissible tolerances shall be increased by 50 percent in lieu of 75 percent.

4.4.1.14 VIBRATION ERROR.- While operating, the indicator shall be subjected to a vibration test in accordance with procedure IV of MIL-E-5272 at amplitudes of 0.009 to 0.011 inch. Permissible tolerances during vibration shall not exceed the room temperature scale error tolerances as specified in 4.4.1.7. Scale error shall be determined at not less than 3 test points for each servo.

4.4.1.15 FREQUENCY RESPONSE.- Frequency response tests will be performed for each servo system within the indicator. Figure 11 shows a typical circuit for obtaining the necessary data. The bandwidth and damping factor can be determined by plotting the ratio of θ-out to θ-in versus frequency (see figure 12). If a servo is less than critically damped the damping ratio is related to the peak on the frequency response curve by an equation for $\zeta < 1$, $M_p = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$ where M_p is the magnitude of the peak and ζ is the damping ratio.

4.4.1.16 SAMPLING PLAN A TESTS.- The indicator shall be subjected to and shall meet the sampling plan A tests of MIL-L-25467.

4.4.1.17 MAGNETIC PROPERTIES.- The magnetic effect of the indicator shall be determined in terms of deflection of a free magnet approximately 1-1/2 inches long, suspended in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted. It shall not cause the free magnet to deflect more than 5° when the indicator is held in various positions on an east-west line with its nearest part 12 inches from the center of the magnet. Tests shall be made with power applied to the indicator.

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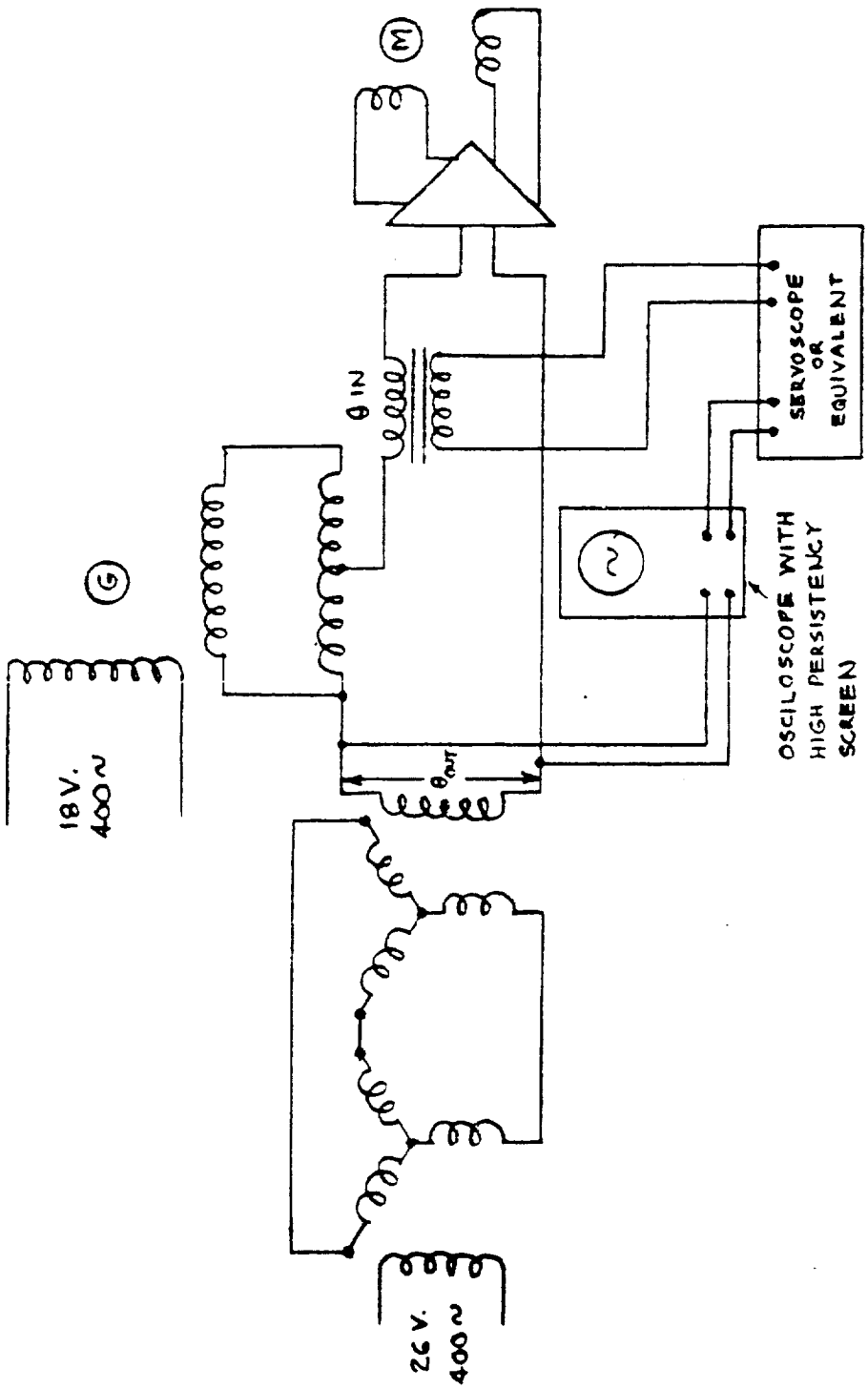


FIGURE 11. RESPONSE TEST CIRCUIT

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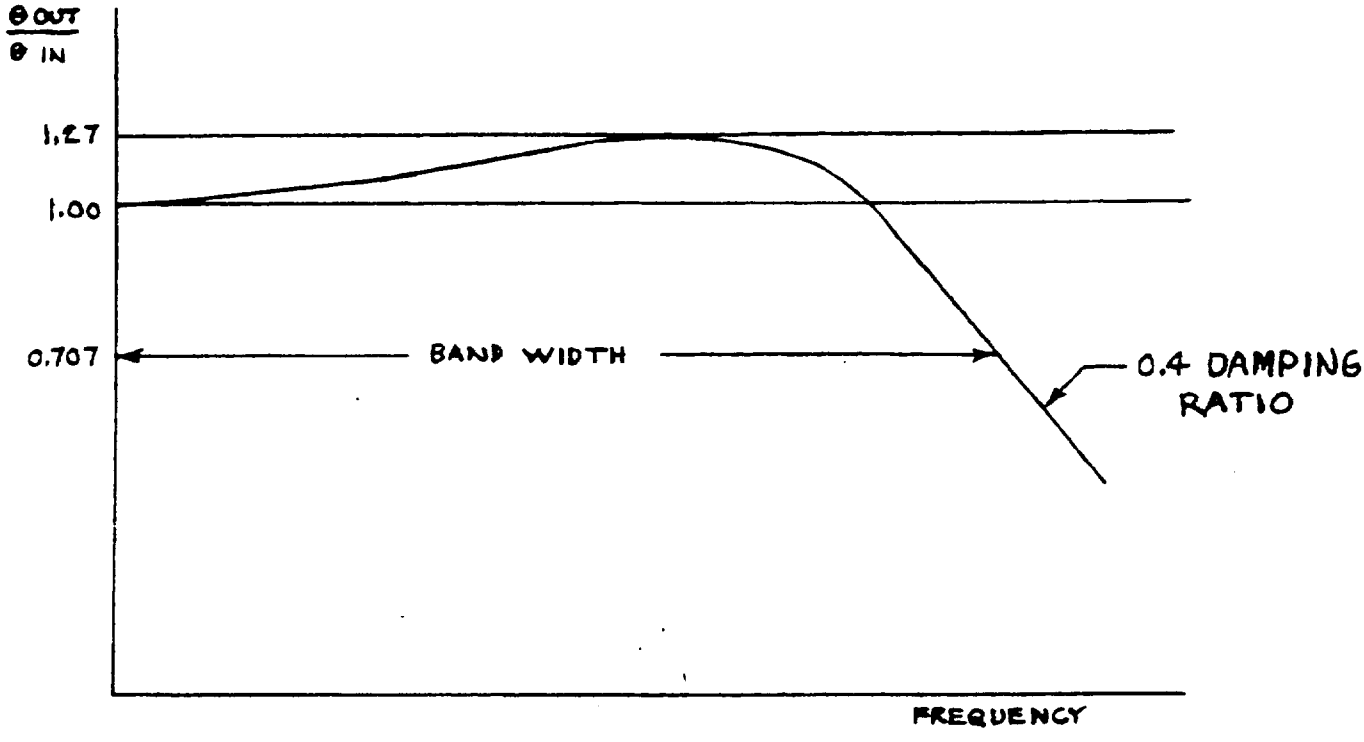


FIGURE 12. BANDWIDTH AND DAMPING CURVE

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4.4.1.18 VIBRATION FAILURE.- While operating, the indicator shall be subjected to a vibration failure test in accordance with procedure V of MIL-E-5272.

4.4.1.19 ALTITUDE.- The indicator shall be subjected to a pressure of 0.820 inches Hg. At this pressure all functions of the indicator shall be capable of operating though full scale. After being returned to sealevel pressure the indicator shall be capable of meeting the requirements of section 3 of this specification.

4.4.1.20 ACCELERATION.- While operating, the indicator shall be subjected to an acceleration test in accordance with procedure II of MIL-E-5272 at 7g.

4.4.1.21 HUMIDITY.- The indicator or a suitable hermetically sealed mockup that exactly duplicates the external portion of the indicator plus the cabin pressure unit shall be subjected to a humidity test in accordance with procedure I of MIL-E-5272 with relative humidity up to 100 percent.

4.4.1.22 FUNGUS.- The indicator or a suitable hermetically sealed mockup that exactly duplicates the external portion of the indicator plus the cabin pressure unit shall be subjected to fungus growth as encountered in the tropical climate in a mold chamber maintaining an internal temperature of 30°C and a relative humidity of 95 percent for a period of 28 days. At the end of this test, the parts shall be visibly examined. Deterioration or corrosion of any internal components which could in any manner prevent the indicator from meeting operational requirements during service life shall be considered failure.

4.4.1.23 SALT SPRAY.- The indicator or a suitable hermetically sealed mockup that exactly duplicates the external portion of the indicator plus the cabin pressure unit shall be subjected to a salt spray test in accordance with MIL-E-5272 for a period of 50 hours.

4.4.1.24 ENDURANCE.- The indicator and amplifier shall be tested for endurance for at least 700 hours of electrical operation with a cycling time for each indicator function of 6 to 7 minutes per cycle (total of at least 6000 cycles per function). The test will begin when rated power is applied. After 4 hours of rated power on, the power shall remain off for a period of 2 hours. This procedure of power-on power-off operation shall be repeated until the 700 hours of actual electrical operation are achieved. After each 50 hours of electrical operation, the indicator shall be subjected to a scale error test of at least 3 test stationd per function. Room tolerances shall apply throughout. The conditions under which

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the system shall be tested are listed in the table below. The table represents the following percentage breakdown of test time.

Room Temp.	64 percent
-55°C	14 percent
+70°C	22 percent

Endurance Test Table

Cycling Period	Number of Hours	Ambient Temperature °C
1	50	25
2	50	70
3	100	25
4	50	-55
5	50	25
6	50	70
7	100	25
8	50	-55
9	100	25
10	50	70
11	50	25

During the fifth cycle the indicator shall be vibrated as specified in procedure IV of MIL-E-5272 at an amplitude of 0.000 to 0.011 inches, and a frequency of 35 cps. The transition from the fourth to fifth cycles shall be completed in the shortest possible time to allow the vibration test to be conducted during temperature change from -55°C to +25°C. In case of malfunction of any kind, the malfunction shall be recorded and the life test interrupted while repairs are being made. The life test shall then be continued from point of interruption. Failure to meet scale error tests shall be considered a malfunction and so recorded. Upon completion of the life test the indicator will again be tested for scale error. If recalibration is required to pass the scale error test an additional malfunction shall be charged against the life test. Upon completion of the test the mean life shall be determined from the following equation.

$$\text{Mean life} = \frac{700N}{M}$$

Where N = number of units under test
M = number of malfunctions
The mean life of the indicator shall be at least 500 hours.

4.4.1.25 SAMPLING PLAN B.- The indicator shall be subject to and shall meet the sampling plan B tests of MIL-L-25467.

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4.4.2 AMPLIFIER TESTS

4.4.2.1 EXAMINATION OF PRODUCT.- Each amplifier shall be inspected for compliance with this specification regarding material and workmanship. Attention shall be given to neatness of wire bundling and wrapping, and the routing of wire bundles to eliminate mechanical and electrical interference.

4.4.2.2 LEAKAGE TEST.- Each module of the amplifier shall be marked indicating that it has been subjected to a leakage test and that the leakage rate of helium as detected by a mass spectrometer did not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.4.2.3 FUNCTIONAL CHECK.- The amplifier shall be excited and all channels shall be used to simultaneously cycle indicator functions at the rate of one cycle per 2 minutes for 2 hours. A cycle shall consist of the excursion from minimum value of the indicator function to maximum value and return to minimum value.

4.4.2.4 INSULATION BREAKDOWN TEST.- This is a destructive test and shall not be done repeatedly. The test shall be performed with the equipment de-energized and with no system connections to the amplifier. Disconnect all grounds from chassis. The application of the following specified sine wave rms voltage at a commercial frequency for a period of 5 seconds between each terminal (designated below) and the case shall not cause an insulation breakdown. The leakage current shall not exceed 4.5 ma.

Test Voltage	Connector	Pin
500	J1201	C
500	J1201	V
500	J1201	B
500	J1201	M
500	J1201	U
500	J1202	F
500	J1202	R
300	J1202	K

4.4.2.5 INSULATION RESISTANCE TEST.- This test is to be performed with the equipment de-energized and with no system connections to the amplifier. Disconnect all grounds from the chassis. The insulation resistance test shall be performed utilizing a test voltage of 500V dc applied for 5 seconds. In all cases the insulation resistance shall not be less than 20 megohms at that voltage. To perform the insulation resistance test apply the test voltage between

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the case and the following test points noting the resistances obtained (using a Megohm Resistance Bridge or "MEGGER"). Readings shall be recorded.

J1201 - Pins C, V, B, M, U
J1202 - Pins F, R, K

NOTE: Replace chassis ground lugs upon completion of this test.

4.4.2.6 CONTINUITY.- This test is to be performed with all external connections removed. The resistance of the circuits shall be as specified below. (Resistance readings shall be ±20 percent.)

Connector	Pin	to Connector	Pin	Resistance (Ohms)
J1202	K	J1202	t	15
J1202	K	J1202	m	100 megs. (min. after capacitor charge)
J1202	K	J1202	L	15
J1202	K	J1202	d	100 megs. (min. after capacitor charge)
J1202	K	J1202	r	15
J1202	K	J1202	k	100 megs. (min. after capacitor charge)
J1202	K	J1202	T	15
J1202	K	J1202	W	100 megs. (min. after capacitor charge)
J1202	h	J1201	B	25K (R1202 full clockwise)
J1202	s	J1202	K	1.0 (max)
J1202	U	J1202	V	5500
J1202	R	J1202	A	15
J1202	B	J1202	P	6100
J1202	K	J1202	E	1.0 (max)
J1202	n	J1202	p	10100
J1202	K	J1202	q	1.0 (max)
J1202	N	J1202	f	5100
J1202	T	J1202	Y	1.0 (max)
J1202	K	J1202	S	1 (max)
J1201	N	J1201	P	1 (max)
J1201	N	J1201	c	1 (max)
J1201	Y	J1202	N	6600
J1201	L	J1201	N	1 (max)
J1201	R	J1201	S	1 (max)
J1201	A	J1201	C	1 (max)
J1201	V	J1201	Z	276
J1202	a	J1201	W	60K to 500K*

*NOTE: Pin J1201 W positive with respect to J1202 a.

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4.4.2.7 RELAY TEST.- With 28V dc applied between pins Z and V on J1201 the following continuity tests shall be made: (No other external connections.)

Connector	Pin	to Connector	Pin	Resistance (Ohms)
J1201	M	J1201	c	1 (max)
J1201	U	J1202	M	60K to 500K*
J1201	Y	J1201	a	1 (max)
J1201	b	J1201	a	100 megohms (min)

*NOTE: Pin M positive with respect to pin U.

With the relay unenergized (28V not applied) the following continuity test shall be made: (No external connections).

Connector	Pin	to Connector	Pin	Resistance (Ohms)
J1201	M	J1201	c	100 megohms (min)
J1201	E	J1202	M	60K to 500K*
J1201	b	J1201	a	820 ±20%
J1201	Y	J1201	a	50K**

*NOTE: Pin M positive with respect to pin E
**NOTE: Pin J1201Y positive with respect to J1201a.

4.4.2.8 VOLTAGE AND CURRENT MEASUREMENTS.- External circuit connections shall be made to the amplifier as shown on figure 13. Apply 115V 400 cps from pins C to S of J1201. This voltage shall be designated as the reference voltage and hereafter shall be referred to as such.

NOTE: Before performing the following tests allow the equipment to warmup for 1 minute.

4.4.2.8.1 AC TEST.- Measure the ac voltages and their corresponding phase angles with respect to the reference voltage at each of the following designated pins:

Pin Designation (J1202)	RMS Voltage Volts	Phase
S to i	9 ±1	0° ±5°
S to j	26 ±3	0° ±5°
S to Y	26 ±3	0° ±5°
S to C	9 ±1	0° ±5°
S to Z	26 ±3	0° ±5°
S to D	9 ±1	0° ±5°
S to e	9 ±1	0° ±5°
S to c	26 ±3	0° ±5°

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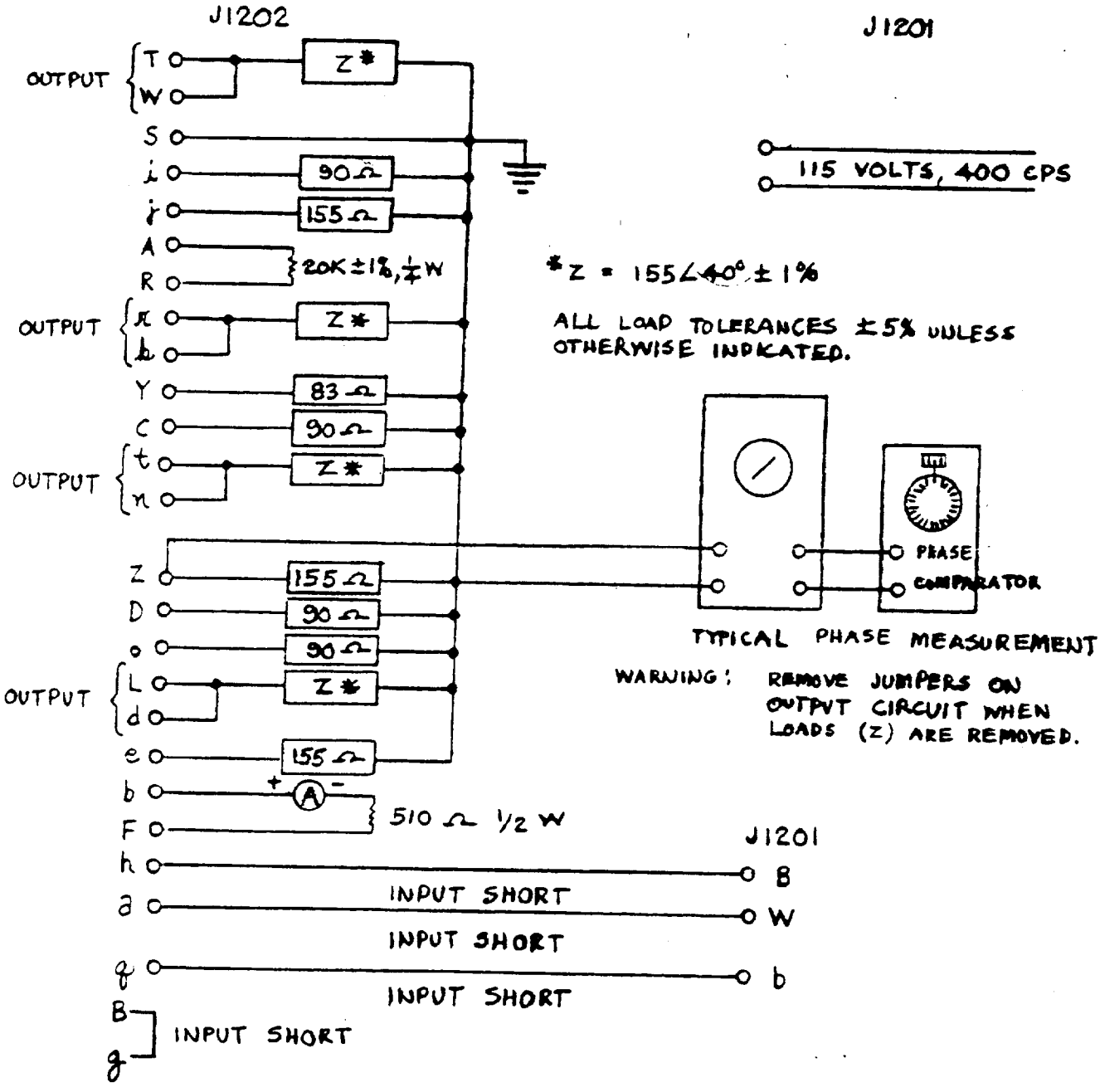


FIGURE 13. AC-DC MEASUREMENTS

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The ratio of the voltages from pins C to S of J1201, V_{CS}, and pins A to R of J1202, V_{AR}, shall be as follows:

$$\frac{V_{AR}}{V_{CS}} = \frac{11.6}{115} \pm 0.1\% = 0.10087 \pm 0.00010$$

The phase of V_{AR} shall be 0° ±18' with respect to V_{CS}.

Voltage ratio test is to be performed after 15 minute warmup.

4.4.2.8.2 POWER FLAG TEST.- The direct current in the 510 ohm resistor connected between J1202 b and J1202F shall be 1.2 ±0.4 ma.

4.4.2.9 POWER CONSUMPTION.- With the amplifier excited with an applied voltage of 115V 400 cps all ac signal inputs shorted and all loads connected as on figure 13 the input current shall be no greater than 0.8 amp, as measured from the 400 cps line.

4.4.2.10 SIGNAL CHANNEL MEASUREMENTS.- The altitude vertical speed amplifier consists of 4 separate signal channels.(altitude, vertical speed, command altitude, and target altitude). Each channel shall be tested separately with signal voltages as called out in its respective figure. The loads shall be connected as on figure 13.

4.4.2.10.1 VERTICAL SPEED (SEE FIGURE 14)

4.4.2.10.1.1 NULL OUTPUT.- With the input shorted and R1201 adjusted for null the total rms voltage across the load shall not exceed 900 mv.

4.4.2.10.1.2 GAIN.- With an input signal of 7 mv ac (0° and 180°) the output shall be 5V ±1.5V.

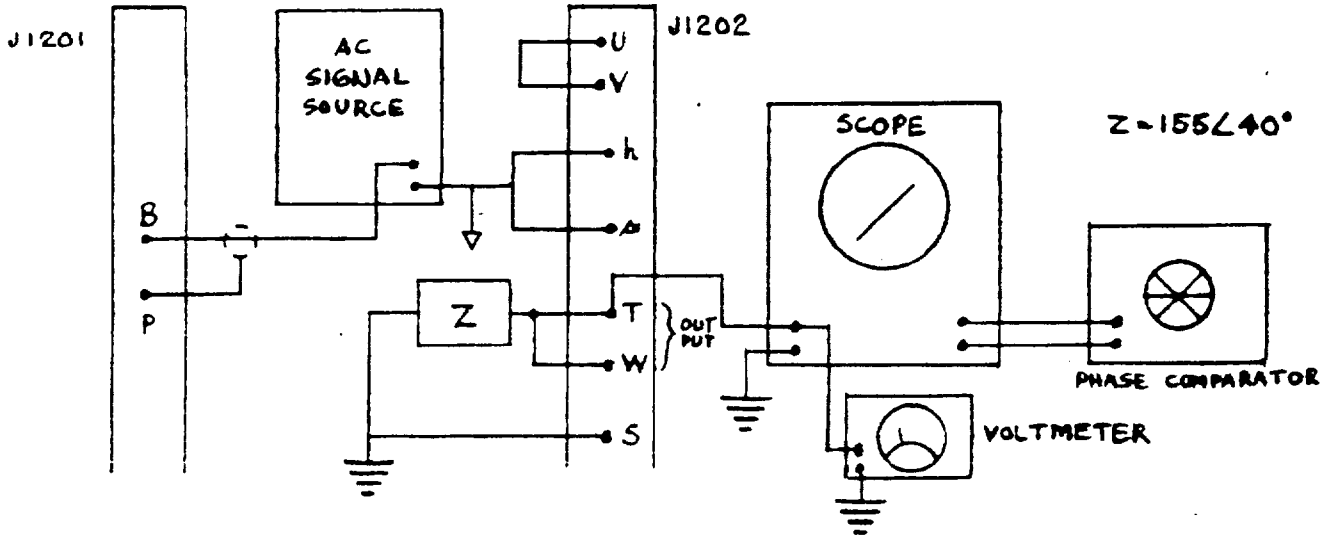
4.4.2.10.1.3 PHASE SHIFT.- For an input signal of 7 mv at an angle of 0° the output voltage shall lead the reference voltage by an angle of 90° ±15°. For an input signal of 7 mv at an angle of 180° the output voltage shall lead the reference by an angle of 270° ±15°.

4.4.2.10.1.4 SATURATION.- The output shall not be less than 13V when the input signal is 100 mv of either phase.

4.4.2.10.2 TARGET ALTITUDE (SEE FIGURE 15)

4.4.2.10.2.1 NULL OUTPUT.- With the dc input shorted the total rms voltage across the load shall not exceed 1.5V.

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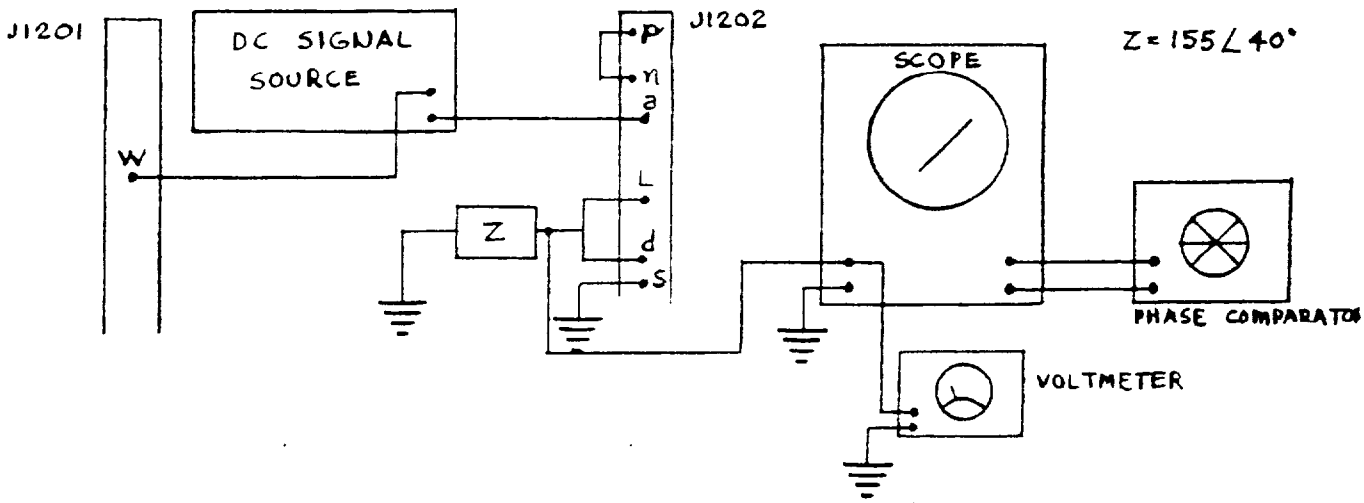


WARNING: REMOVE JUMPER ON OUTPUT CIRCUIT WHEN LOAD IS REMOVED.

NOTE: SHORT INPUTS OF AMPLIFIER NOT UNDER TEST, AND KEEP THE LOADS ON THE OTHER AMPLIFIER OUTPUTS IN ACCORDANCE WITH FIGURE 13

FIGURE 14. VERTICAL SPEED - NULL, GAIN & PHASE SHIFT

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WARNING: REMOVE JUMPER ON OUTPUT CIRCUIT WHEN LOAD IS REMOVED.

NOTE: SHORT INPUTS OF AMPLIFIER NOT UNDER TEST, AND KEEP THE LOADS ON THE OTHER AMPLIFIER OUTPUTS IN ACCORDANCE WITH FIGURE 13

FIGURE 15. TARGET ALTITUDE - NULL, GAIN & PHASE SHIFT

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4.4.2.10.2.2 GAIN.- With an input of ±80 mv dc the output shall be 7.0 ±2.5V.

4.4.2.10.2.3 PHASE SHIFT.- For an input signal of 80 mv dc J1201 W positive the output voltage shall lead the reference voltage by 90 ±20°. With an input signal of 80 mv dc, J1201 a positive, the output signal shall lead the reference voltage by 270° ±20°.

4.4.2.10.2.4 SATURATION.- The output shall not be less than 13V when the input signal is ±500 mv dc.

4.4.2.10.3 COMMAND ALTITUDE (See figure 16).- This test consists of two parts. For the first part of the test apply 28V dc to pins Z and V of J1201 use a dc signal source for the input and take measurements (see figure 16). For the second part use an ac signal source for the input and without applying the 28V dc take measurements (see figure 17).

4.4.2.10.3.1 NULL DC INPUT.- With the dc input shorted the total rms voltage across the load shallnot exceed 1.5V.

4.4.2.10.3.2 GAIN DC INPUT.- With an input of ±80 mv dc the output shall be 7.0 ±2.5V.

4.4.2.10.3.3 PHASE SHIFT(DC INPUT).- For an input signal of 80 mv dc, J1202M, positive, the output voltage shall lead the reference by 90° ±20°. With an input signal of 80mv dc, J1202U, positive, the output signal shall lead the reference voltage by 270° ±20°.

4.4.2.10.3.4 SATURATION (DC INPUT).- The output shall not be less than 13V when the input signal is ±500 mv dc.

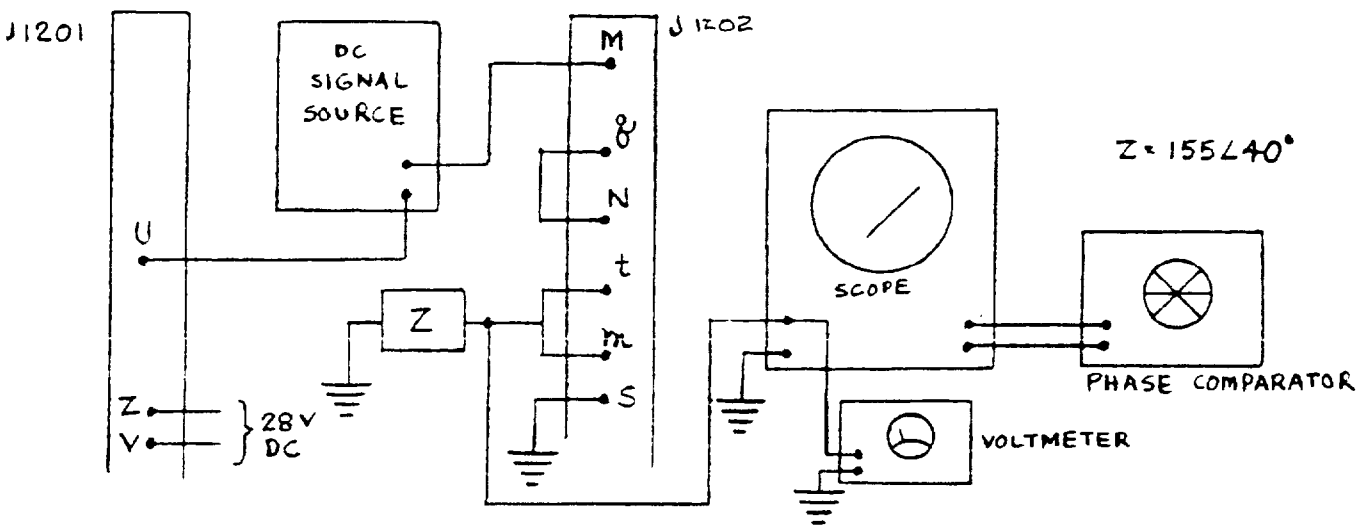
4.4.2.10.3.5 NULL(AC INPUT).- With the ac input shorted the total rms voltage across the load shall not exceed 900 mv.

4.4.2.10.3.6 GAIN (AC INPUT).- With an ac input signal of 10 mv (0° and 180°) the output shall be 5V ±1.5 V.

4.4.2.10.3.7 PHASE SHIFT (AC INPUT).- For an input signal of 10 mv at an angle of 0° the output shall lead the reference voltage by 270° ±15°. For an input signal of 10 mv at an angle of 180° the output voltage shall lead the reference voltage by 90° ±15°.

4.4.2.10.3.8 SATURATION (AC INPUT).- The output shall not be less than 13V when the input signal is 200 mv of either phase.

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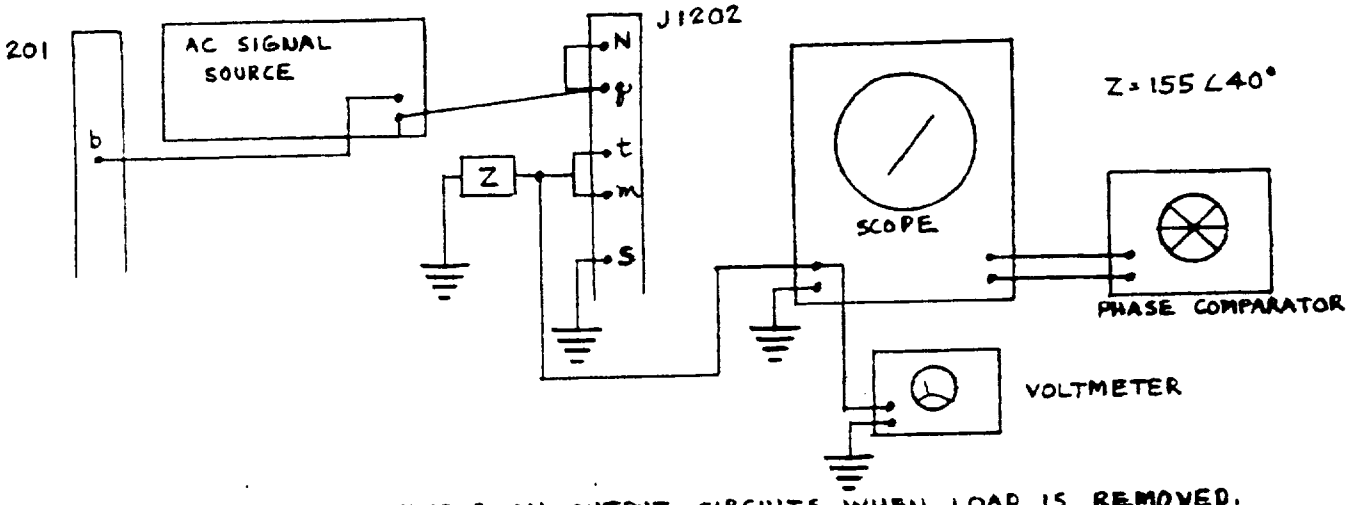


WARNING: REMOVE JUMPER ON OUTPUT CIRCUITS WHE LOAD IS REMOVED.

NOTE: SHORT INPUTS OF AMPLIFIER NOT UNDER TEST, AND KEEP THE LOADS ON THE OTHER AMPLIFIER OUTPUTS IN ACCORDANCE WITH FIGURE 13

FIGURE 16. COMMAND ALTITUDE (AUTOMATIC MODE)

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WARNING: REMOVE JUMPER ON OUTPUT CIRCUITS WHEN LOAD IS REMOVED.

NOTE: SHORT INPUTS OF AMPLIFIER NOT UNDER TEST, AND KEEP THE LOADS ON THE OTHER AMPLIFIER OUTPUTS IN ACCORDANCE WITH FIGURE 13.

FIGURE 17 COMMAND ALTITUDE (MANUAL MODE)

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4.4.2.10.4 ALTITUDE (SEE FIGURE 18)

4.4.2.10.4.1 NULL.- With the fine input shorted the total rms voltage across the load shall not exceed 900 mv.

4.4.2.10.4.2 GAIN.- The gain of the amplifier with the signal applied to the coarse and fine input terminals shall be as follows:

a. Fine input.- With 12 mv (0° and 180° phase) applied to the fine input terminals the output shall be 5V ±1.5V.

b. Coarse input.- Short fine input for this test only. With 150 mv (0° and 180° phase) applied to the coarse input terminal the output shall not exceed 1.5V. With 500 mv (0° and 180° phase) applied to the coarse input terminals the output shall be 5V minimum.

4.4.2.10.4.3 PHASE SHIFT.- For an input signal of 12mv applied to fine input terminals at an angle of 0° the output shall lead the excitation voltage by 270° ±15°. For an input signal of 12 mv at an angle of 180° the output shall lead the excitation voltage by 90° ±15°.

4.4.2.10.4.4 SATURATION.- The output shall not be less than 13V when the fine input is 200 mv of either phase.

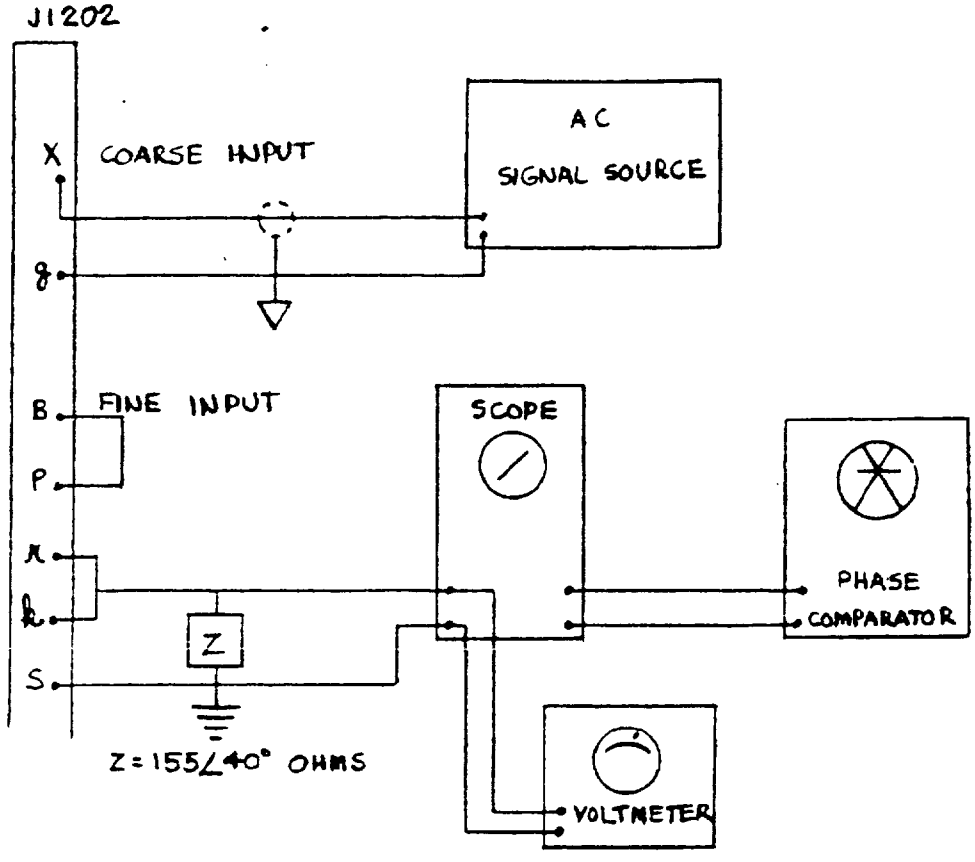
4.4.2.11 LOW TEMPERATURE OPERATION.- The amplifier shall be subjected to a temperature of -55°C for a period of 4 hours. After exposure and with the ambient temperature maintained at -55°C and following a 3 minute warmup period, with power applied, the item shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4 except the following tolerances shall apply:

Null: 1.5V maximum
Gain: AC 5V ±2V DC 7V ±3.5V
Phase Shift: 90° ±30° and 270° ±30°
Saturation: 13V minimum.

4.4.2.12 HIGH TEMPERATURE OPERATION.- The amplifier shall be subjected to a temperature of +71°C for a period of 4 hours. After exposure and with the temperatures maintained at +71°C the item shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4 except the following tolerances shall apply:

Null: 1.5V maximum with ac input shorted and 2V maximum with dc input shorted.
Gain: AC 5V ±2V, DC 7V ±3.5V
Phase Shift: 90° ±30° and 270° ±30°
Saturation: 13V minimum.

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WARNING: REMOVE JUMPER ON OUTPUT CIRCUITS WHEN LOADS ARE REMOVED.
NOTE: SHORT INPUTS OF AMPLIFIER NOT UNDER TEST, AND KEEP THE LOADS IN ACCORDANCE WITH FIGURE 13

FIGURE 18. ALTITUDE SERVO-NULL, GAIN & PHASE SHIFT

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4.4.2.13 VIBRATION.- With the unit operating tests shall be performed without shockmounts and in accordance with MIL-E-5272 procedure I except that the test schedule shall be 0.4 inch double amplitude from 5 to 7 cps and a maximum of 1g acceleration from 7 to 500 cps. The cycling portion of procedure I only shall be conducted. The unit shall suffer no physical damage as a result of this test. During this test each channel of the amplifier shall be tested and shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4.

4.4.2.14 EXTREME TEMPERATURE EXPOSURE.- After alternate exposure to the temperature of -55°C and +71°C for periods of 24 hours each and a delay of 3 hours at room temperature following completion of this exposure the unit shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4.

4.4.2.15 TEMPERATURE-ALTITUDE TESTS.- The temperature-altitude shall be conducted in accordance with the conditions as specified on figure 8. At any combination of temperature-altitude as specified on figure 8 curves A, B, and C the amplifier shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4 except the following tolerances shall apply:

Null: 1.5V maximum with ac input shorted and 2V with dc input shorted
Gain: AC 5V ±2V, DC 7V ±3.5V
Phase Shift: 90° ±30° and 270° ±30°
Saturation: 13V minimum.

4.4.2.16 SHOCK.- The unit shall be subjected to the tests as specified in MIL-E-5272, procedure II. The unit shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4. The unit shall suffer no mechanical failures due to the applied shocks.

4.4.2.17 ACCELERATION.- The unit shall be subjected to the tests as specified in MIL-E-5272 procedure II of 10g. During the test each channel of the unit shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4.

4.4.2.18 HUMIDITY.- The unit shall be subjected to the tests as specified in MIL-E-5272 procedure I. At the completion of the tests each channel of the unit shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4.

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4.4.2.19 RADIO INTERFERENCE.- The unit shall be subjected to the requirements of MIL-E-5400 except that the upper frequency limit of the radiated noise requirements shall be 1000 Mc/sec. The conducted noise requirements of MIL-I-26600 shall not apply to the frequency range nor shall they apply to those conductors which interconnect components.

4.4.2.20 SALT SPRAY.- The unit shall be subjected to the salt spray test of MIL-E-5272. At the completion of the test the unit shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4.

4.4.2.21 FUNGUS RESISTANCE TEST.- The unit shall be subjected to the fungus resistance test of MIL-E-5272. At the completion of the test the unit shall meet the requirements of 4.4.2.7 through 4.4.2.9 and 4.4.2.10.1 through 4.4.2.10.4.

4.4.2.22 MAGNETIC PROPERTY.- The amplifier shall be magnetically shielded. When held in any position such that the nearest portion of the amplifier is a distance of 12 inches from the pivot of the card of an aircraft magnetic compass in a horizontal magnetic field having a strength of 0.18 oersted, the amplifier shall not cause a change in reading of the compass card to exceed 5°.

4.4.2.23 ENDURANCE.- The amplifier shall be subjected to an endurance test in conjunction with an indicator as specified in 4.4.1.24.

4.5. PREPRODUCTION TESTING

4.5.1 PREPRODUCTION TEST SAMPLE TESTED BY THE CONTRACTOR.- The contractor shall subject 3 indicators and amplifiers to the preproduction test specified in 4.5.4.

4.5.2 PREPRODUCTION TEST REPORT.- After the contractor completes the preproduction tests, he shall prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.5.3 PREPRODUCTION TEST SAMPLE FOR THE PROCURING ACTIVITY.- Along with the preproduction test report, the contractor shall submit sample(s) to the procuring activity who will use it (them)

a. For a review of the mechanical construction of the product.

b. To perform any tests, included in the specification, after reviewing the contractor's test report.

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4.5.4 PREPRODUCTION TESTS.- The preproduction tests shall consist of all the tests specified herein.

5. PREPARATION FOR DELIVERY

5.1 GENERAL.- The indicator and amplifier shall be prepared for delivery in accordance with MIL-P-7936. Preservation shall be in accordance with method III of MIL-P-116.

5.2 MARKING OF SHIPMENT.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature for the items comprising the A/A24G-7 amplifier-indicator group shall be as follows:

Indicator, Altitude-Vertical Speed AAU-1/A24G-7 *
or
Amplifier, Electronic Control ASK-6/A24G-7 *

* Applicable nomenclature to be entered by the contractor.

5.2.1 REINSPECTION DATE.- The reinspection date markings shall be as specified by the procuring activity.

6. NOTES

6.1 INTENDED USE.- The indicator and amplifier covered in this specification is intended for use in the cockpit of high performance weapon systems.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, date, and number of this specification.

b. Level of packaging and packing desired

c. Number of preproduction samples and point of testing

d. Whether sampling plan B tests are to be conducted

e. Whether procurement covers the indicator or amplifier, or both.

f. Reinspection date markings.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8878

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